

WHY HAS FACTOR INVESTING FAILED?: THE ROLE OF SPECIFICATION ERRORS

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KEY TAKEAWAYS

- The econometric cannon neglects the existence of colliders (variables influenced by both treatment and outcome), inducing factor researchers to select misspecified models.
- Adding a collider to a factor model’s specification introduces biases in the estimation of the factor’s beta. When the factor’s beta is estimated with the wrong sign, the investor buys securities that should have been sold and vice versa, leading to systematic losses.
- Practical tools such as directed acyclic graphs, causal discovery algorithms, subject-matter knowledge, and targeted robustness checks help researchers identify and mitigate the influence of colliders on causal estimates in observational studies.

ABSTRACT

In factor investing, *p*-hacking is a well-understood cause of false positives. A far less studied cause is factor model specification choices encouraged by the current econometric canon. We prove that specification errors cause factor strategies to underperform and potentially yield systematic losses, even if *all* risk premia remain constant and are estimated with the correct sign. Unlike the *p*-hacking-driven factor zoo, where noise is mistaken for signal through repeated testing, we identify a distinct phenomenon—the factor mirage—in which canonical econometric practices systematically reward misspecified models that appear statistically strong but are structurally flawed. We show that these practices, especially over-controlling for colliders, increase the likelihood of few-shot *p*-hacking and adverse outcomes. The implication is that specification errors are a more insidious and underappreciated threat to investors than previously recognized. To our knowledge, this is the first study to connect factor model selection, collider bias, underperformance, and systematic losses through a unified causal framework. These findings challenge the scientific credibility and long-term viability of the current associational (non-causal) multi-trillion-dollar factor investing industry. To address these risks, we propose adjustments to the econometric canon, informed by recent advances in machine learning and causal inference.

Keywords: Causal inference, causal discovery, confounder, collider, factor investing, *p*-hacking, underperformance, systematic losses.

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1. INTRODUCTION

Factor investing can be defined as the investment approach that aims to monetize the exposure to measurable risk characteristics (called “factors”) that presumably explain differences in the performance of a set of securities. Its origins can be traced back to the seminal work of Fama and MacBeth [1973] and Schipper and Thompson [1981], among others. Since then, thousands of academic papers have claimed the discovery of hundreds of investment factors (Harvey et al. [2016]), propelling the growth of a multi-trillion-dollar industry. In 2019, J.P. Morgan estimated that over USD 2.5 trillion (more than 20 percent of the US equity market capitalization) was managed by quant-style funds (Neuberger Berman [2019]). BlackRock estimates that the factor investing industry managed USD 1.9 trillion in 2017, and it projected that amount would grow to USD 3.4 trillion by 2022 (BlackRock [2017]).



Exhibit 1 - Performance of US multi-factor equity strategies. Source: Bloomberg

Unfortunately for investors, these academically-endorsed investment products have failed to perform as expected. Exhibit 1 plots the performance of one of the broadest factor investing indices, the Bloomberg – Goldman Sachs Asset Management US Equity Multi-Factor Index (BBG code: BGSUSEMF <Index>). This Multi-Factor index tracks the long/short performance of the momentum, value, quality, and low-risk factors in U.S. stocks (Bloomberg [2021]). Its annualized Sharpe ratio from May 2, 2007 (the index’s inception date) to July 19, 2024 (this paper’s completion date) has been 0.17 ($t\text{-stat}=0.69$, $p\text{-value}=0.25$), and the average annualized excess return has been 0.62%. This performance does not include: (a) transaction costs; (b) market impact of order execution; (c) cost of borrowing stocks for shorting positions; (d) management and incentive fees. Including those costs would yield a negative total investment return. After almost 17 years of out-of-sample performance, factor investing’s Sharpe ratio is statistically insignificant at any reasonable confidence level.

The primary goal of this paper is to elucidate why the performance of factor investing strategies has disappointed, and what can be done to correct this situation. While past critiques have focused on multiple testing and p -hacking—the statistical overfitting that populates the so-called “factor zoo” (Cochrane [2011], Bailey et al. [2014], Fabozzi and López de Prado [2018], Harvey and Liu [2020])—we highlight a subtler but equally damaging phenomenon: the factor mirage. Unlike p -hacked strategies, a factor mirage arises not from data mining but from specification errors embedded in canonical econometric practices, particularly the inclusion of colliders or the omission of confounders. These misspecified models may exhibit strong explanatory power and seemingly robust statistical properties, while in fact delivering biased estimates and potentially systematic losses—even when all risk premia remain constant and correctly signed. Because these flawed models comply with widely accepted econometric procedures, they have passed peer review in top journals and made their way into investment products. Unless the canon is updated to reflect causal principles, factor investors have no reason to expect better performance in the future. This paper exposes the mechanism behind the mirage and offers a research protocol to prevent it.

The rest of the paper is organized as follows: Section 2 outlines the paper’s contributions to the literature. Section 3 shows how factor model misspecification cause underperformance and systematic losses in factor investing. Section 4 proves that models that over-control for a collider always exhibit a higher R-squared, and under some conditions lower p -values. Section 5 exposes the misguided standard econometric practice of choosing a model’s specification on the basis of its higher explanatory power. Section 6 explains the causal mechanism that can explain the underperformance and systematic losses delivered by factor investing. Section 7 proposes a research protocol for the discovery of causal factors, with detailed recommendations for practitioners and academics. Section 8 summarizes our conclusions. The Appendix contains mathematical proofs of propositions used throughout the paper, and a glossary of terms.

2. INNOVATIONS

When a factor is itself a portfolio, and therefore it is tradable, its risk premium can be simply estimated as the average excess return of that portfolio. The best-known example of a tradable factor is the market portfolio. In most cases, however, economic theories point to non-tradable factors, such as liquidity, and many variables associated with macroeconomic regimes, intangibles, or the business cycle. To estimate the risk premium of non-tradable factors, researchers typically implement a two-pass estimator (Jensen et al. [1972], and Fama and MacBeth [1973]). In the first pass, time series regressions estimate the exposures of securities to a factor, while controlling for all other risks to which the securities are exposed. In the second pass, cross-sectional regressions estimate the risk premium associated with exposure to the factor of interest, while controlling for all other exposures. All two-pass estimators are afflicted by two potential specification errors: (a) missing a confounder (also known as the omitted variable problem); and (b) including a collider (i.e., a variable that is dependent on both the right- and left-hand side variables in the regression).¹ These estimators suffer from errors-in-variables (EIV) bias, particularly in the second pass, as noted by Shanken [1992], who proposes a correction based on the covariance of first-pass estimates. To address the specification error due to a missing confounder, Giglio and Xiu [2021] have proposed a three-pass method to estimate the risk premium of an observable factor, which is

¹ Collider bias should not be mistaken with simultaneous equations bias, i.e. the bias that arises from having two or more variables determined jointly within a system of equations (a form of cyclical causality).

valid even when not all factors in the model are specified or observed. Virtually all factor investing papers and products predate the development of methods resilient to omitted variables or EIV bias, and hence likely suffer from multiple sources of estimation error. The second type of specification error, due to the inclusion of a collider, remains an unsolved investment problem, as no econometric procedure is known to produce unbiased estimates of risk premia in that scenario.² This paper studies the effects of these two specification errors on the performance of factor investing strategies, and it proposes ways to address those errors.

Our contributions to the financial economics and econometrics literatures can be grouped in four themes. First, we show that the assumption of correct model specification is a defining feature that sets factor investing apart from other types of investment strategies, such as forecasting strategies. We study the effect of specification errors on the performance of factor and forecasting strategies. We prove that both types of strategies can underperform and yield systematic losses even if the risk premia remain constant and are estimated with the correct sign.

Second, several authors have argued that the erratic performance of factor investing strategies is due to time-varying risk premia (see, for example, Evans [1994], Anderson [2011], and Cochrane [2011]). Our findings show that this erratic performance is better explained (in simpler, more parsimonious terms) by specification errors, without requiring changes in the market’s compensation for risk. The risk premia can remain constant, and specification errors will still cause factor investing strategies to perform erratically.

Third, we prove sufficient conditions for a misspecified factor model to exhibit a higher R-squared and lower p -values than the correctly specified model, despite producing biased parameter estimates. Kan et al. [2013] discuss confounder bias in the context of model misspecification, however they do not account for collider bias, which is the more relevant form of misspecification in explaining R-squared inflation or p -value deflation.

Fourth, we describe a causal mechanism whereby the econometric canon enables p -hacking within a few trials, by favoring over-controlled (and possibly money-losing) models with higher R-squared and lower p -values than the correctly specified model. The implication is that the proliferation of spurious factors due to specification errors—what we call the “factor mirage”—may be more harmful to investors than the data-mining-fueled “factor zoo” (Cochrane [2011]), because canon-compliant, albeit misspecified, factor models can yield systematic losses even if the risk premia remain constant and are estimated with the correct sign.

To our knowledge, this is the first study that connects factor model selection practices, specification errors, underperformance and systematic losses through a causal mechanism.

3. THE EFFECT OF SPECIFICATION ERRORS ON STRATEGY PERFORMANCE

Factor strategies decide whether a security i is bought or sold based on the value of $\hat{\beta}X_{i,t}$ at time t , where $X_{i,t}$ measures i ’s exposure at time t to a risk characteristic that the market rewards with

² Note that the inclusion of a collider is different from the inclusion of an irrelevant factor, because irrelevant factors are uncorrelated with the test asset returns. See Gospodinov et al. [2014] for a discussion of irrelevant factors.

an estimated linear premium $\hat{\beta}$, as determined by a cross-sectional factor model with $|\hat{\beta}| \gg 0$.³ The goal of the analysis is not to predict the future return of i , but to establish whether the market rewards on average holding securities with exposure X (a common non-diversifiable rewarded risk).⁴ Estimating the risk premium β without bias requires that the factor model is correctly specified. For instance, a value strategy may buy a security i when its book-to-price ratio is high, and hedge other risks by selling a security j whose book-to-price ratio is low, where i and j are similar in everything but their different exposure to book-to-price ratio. Provided that the factor model is correctly specified and β 's estimate is unbiased and sufficiently positive, the strategy profits from i 's outperformance over j . However, should the factor model be misspecified, the parameter estimates will be biased ($E[\hat{\beta}|X] \neq \beta$), the strategy will have exposures other than X , and the investor will likely experience underperformance, including potentially systematic losses.

In contrast, forecasting strategies decide whether a security i is bought or sold based on the forecasted return of i at time t . These strategies do not attempt to profit from holding securities exposed to a rewarded risk characteristic while hedging other exposures. Instead, forecasting strategies attempt to monetize associations, regardless of their source. For instance, a pairs trading strategy (a type of statistical arbitrage) may buy or sell the spread i between two cointegrated securities, where $\hat{\epsilon}_{i,t}$ measures the value of the residual of an error-correction model, a type of time series analysis (López de Prado and Leinweber [2012]). That strategy profits from the forecast that $\hat{\epsilon}_{i,t}$ will converge to zero over time, even if the forecasting model is misspecified and the parameter estimates are biased.

The above explanation highlights that the assumption of correct model specification sets factor investing strategies apart, as other investment styles do not make decisions based on premia attributed to controlled risk characteristics. The violation of this assumption (correct model specification) has practical consequences in terms of profitability that factor academics and practitioners need to understand.

Factor model misspecification has two primary consequences for investors: Type-1 errors (false strategies) and type-2 errors (missed strategies). Out of the type-1 errors, some of them result in systematic losses (a factor strategy with an expected negative return). The next section studies in detail the effect of factor model misspecification on strategy performance.

3.1. MISSING A CONFOUNDER

Consider a data-generating process with equations:

$$\begin{aligned} X &= Z\delta + v \\ Y &= X\beta + Z\gamma + u \end{aligned} \tag{1}$$

³ The factor investing and econometrics literatures use different notations. Some factor investing authors use $\beta_{i,t}$ to denote the exposure of security i at time t to a rewarded risk characteristic. The exposure $\beta_{i,t}$ could be measured (e.g., book-to-price ratio) or estimated through a time series regression. In both cases, the premium λ_t associated with $\beta_{i,t}$ is estimated through a cross-sectional regression. See Ferson [2019, chapter 22] for a detailed explanation. In this paper we use the prevailing econometric notation, as it is more relevant to misspecification problems.

⁴ Factor investing models do not typically lag the explanatory variables, so the explanatory and explained variables are observed at the same time. For example, see the equations in Tables 3-11 of Fama and French [1993], and equations (4)-(6) in Fama and French [2015]. This modelling choice makes it possible for reverse causation to occur.

where $\gamma \neq 0$, $\delta \neq 0$, and variables (u, v, Z) are independent and identically distributed as a standard Normal, $(u, v, Z) \sim N[0, I]$.⁵ The causal graph for this process is displayed in Exhibit 2.⁶ This is the graph that represents through nodes all the variables involved in the process, and through arrows the direction of dependences. In the language of causal inference, Z is called a confounder, because it influences both the treatment (X) and the outcome (Y), thus obfuscating the true effect of X on Y .⁷ The arrow in green denotes the causal path from X to Y , and the arrows in red denote a non-causal path from X to Y through Z , i.e. $X \leftarrow Z \rightarrow Y$. In presence of a confounder, this is also called a backdoor path, because it has an arrow pointing into the cause (X).

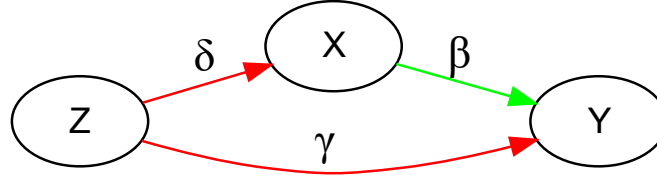


Exhibit 2 – Variable Z as confounder

The data-generating process is unknown to observers. With proper knowledge of the causal graph, observers would estimate the effect of X on Y by fitting the equation $Y = X\beta + Z\gamma + \varepsilon$. Controlling for confounder Z blocks the backdoor path, thus achieving that $E[Y|X, Z] = E[\hat{Y}|X, Z]$, i.e. the true value of Y (from the data-generating process) is on average equal to the value of Y expected by observers, and the parameters are estimated without bias, $E[\hat{\beta}|X, Z] = \beta$, $E[\hat{\gamma}|X, Z] = \gamma$.

Suppose that observers incorrectly attempt to estimate the causal effect of X on Y by fitting the equation $Y = X\beta + \varepsilon$ on a sample produced by the process. Then, the backdoor path remains unblocked, and the expected value of the estimated β is,

$$E[\hat{\beta}|X] = \beta + \frac{\gamma\delta}{1 + \delta^2} \quad (2)$$

See Section A.1 for a proof of the above statement. For appropriate values of (δ, γ) , the confounder bias $\frac{\gamma\delta}{1 + \delta^2}$ can be so large that it flips the sign of the estimated β (a phenomenon known as Simpson's paradox, see Pearl [2014]).⁸ In this case, observers expect that the value of Y is

⁵ Serial independence, linearity, stationarity and Normality are highly idealized assumptions commonly adopted in the factor investing literature. One goal of this paper is to prove that, even under this ideal scenario, factor strategies are structurally flawed and will not perform as designed.

⁶ In this paper, we focus on causal graphs that are acyclic. Acyclicity allows us to represent joint distributions as the product of conditional probabilities between ancestors and descendants only. This choice does not limit the analysis to less realistic scenarios, because acyclicity can usually be imposed through: (a) redefinition of variables, e.g. adding lagged variables as causes; and (b) targeted sampling frequency, such that the effect on Y is measured before feedback loops come into play.

⁷ For a general introduction to causal inference, causal discovery and do-calculus, see Neal [2020], Glymour et al. [2019], Pearl [2009], Pearl et al. [2016]. For an introduction to causal inference in finance, see López de Prado [2023a].

⁸ This bias in the parameter estimation is not in contradiction with the least-squares property whereby $E[Y - E[\hat{Y}|X]] = 0$.

$$E[\hat{Y}|X] = XE[\hat{\beta}|X] \quad (3)$$

whereas the true value (from the data-generating process) of Y has expectations

$$\begin{aligned} E[Y|X] &= X\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right) \\ E[Y|Z] &= Z(\gamma + \beta\delta) \\ E[Y|X, Z] &= X\beta + Z\gamma \end{aligned} \quad (4)$$

From an investment standpoint, variables X and Z are factor exposures, because both are a cause of Y . However, not all system parameters are risk premia. This can be seen in the correctly specified model, $E[\hat{Y}|X, Z] = XE[\hat{\beta}|X, Z] + ZE[\hat{\gamma}|X, Z] = X\beta + Z\gamma = E[Y|X, Z]$, thus Y is only influenced by two parameters (β, γ) . Therefore, (β, γ) are risk premia, and δ is not a risk premium. Parameter δ is a confounding effect that may bias the estimate of risk premium β (note that the bias is zero when $\delta = 0$). Next, we study the implications of under-controlling for a confounder on the performance of factor strategies and forecasting strategies, when system parameters (β, γ, δ) do not change, hence all correlations remain constant.

3.1.1. FACTOR STRATEGIES

Consider an investor who wishes to harvest the risk premia offered by securities with returns Y , where the data-generating process is the one described in Section 3.1. With the correctly specified factor model, the estimates of the risk premia are unbiased, and the factor strategy controls for X and confounder Z . A factor strategy that sizes positions by $E[\hat{Y}|X, Z]$ produces expected returns

$$\begin{aligned} E[YE[\hat{Y}|X, Z]|X, Z] &= E[Y(X\beta + Z\gamma)|X, Z] \\ &= E[Y|X, Z](X\beta + Z\gamma) \\ &= (X\beta + Z\gamma)^2 \geq 0 \end{aligned} \quad (5)$$

The above equation shows that the factor strategy based on correctly specified factor model extracts the risk premia β and γ , as it was designed to do. The equation excludes the confounding effect δ , because controlling for Z blocks the backdoor path. Alternatively, an investor with a correctly specified model may decide to harvest only X , for instance if Z is a confounding but non-investable variable. In this case, a factor strategy that sizes positions by $E[\hat{Y}|X]$ produces expected returns

$$\begin{aligned} E[YE[\hat{Y}|X]|X] &= E\left[(X\beta + Z\gamma + u)X\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right)|X\right] \\ &= (\beta X + \gamma E[Z|X] + E[u|X])X\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right) \\ &= X\left(\beta X + \gamma \frac{\delta}{1 + \delta^2} X + 0\right)X\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right) \\ &= X^2\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right)^2 \geq 0 \end{aligned} \quad (6)$$

Now the confounding effect δ is also present, and the strategy failed to produce as expected return $(X\beta)^2$. This is an important result, because it highlights that using a factor model with the correct specification is not enough: the investment strategy must control for the confounder when choosing securities and sizing positions. It also shows that the standard industry practice of creating single-factor funds is misguided, as those funds are likely exposed to the confounding effect δ . Furthermore, pooling multiple single-factor strategies into one fund does not eliminate this confounding effect δ .

With the incorrect model specification $Y = X\beta + \varepsilon$, the investor aims to harvest X , but without controlling for Z 's confounding effect on Y . If positions are sized by $E[\hat{Y}|X]$, then the expected returns of this strategy are

$$\begin{aligned} E[YE[\hat{Y}|X]|X] &= E\left[YX\left(\beta + \frac{\gamma\delta}{1+\delta^2}\right)|X\right] \\ &= E[Y|X]\left(\beta + \frac{\gamma\delta}{1+\delta^2}\right)X \\ &= X^2\left(\beta + \frac{\gamma\delta}{1+\delta^2}\right)^2 \geq 0 \end{aligned} \tag{7}$$

Again, the confounding effect δ is present, and the strategy failed to produce expected return $(X\beta)^2$. A factor strategy that under-controls for a confounder produces the same performance as a factor strategy that used the correct model specification but failed to harvest the confounder.

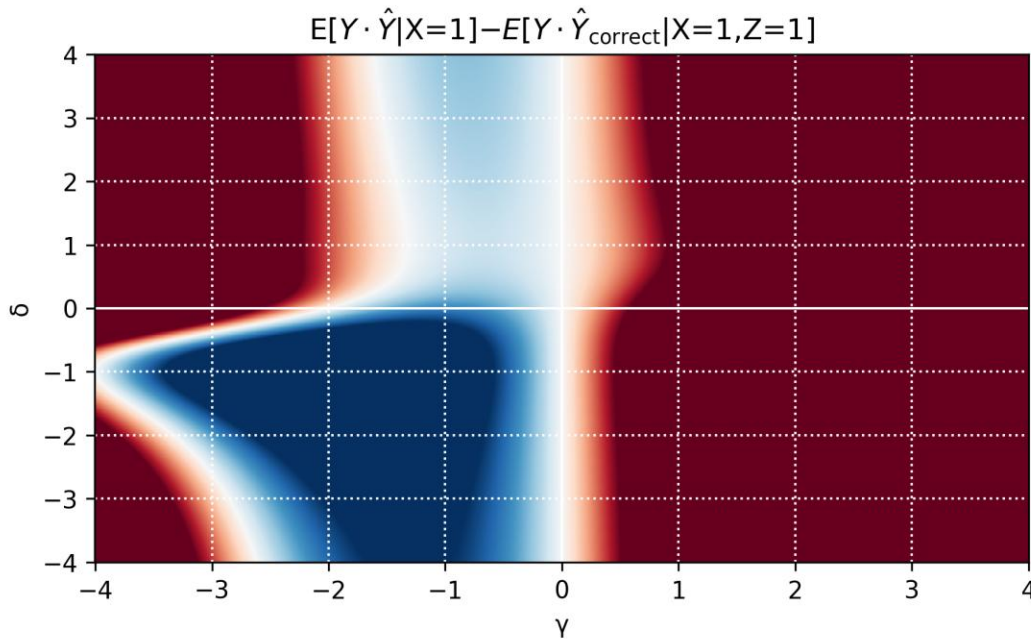


Exhibit 3 – Underperformance (red) and outperformance (blue) of a factor strategy that under-controls for a confounder, where $X = 1, Z = 1, \beta = 1$ and different combinations of (γ, δ)

The conclusion from the above equations is that under-controlling for a confounder causes factor strategies to underperform, for two reasons: (a) the correctly specified model allows the strategy to harvest two factors, whereas the under-controlled model limits the harvesting to only one; and (b) the non-causal association created by the open backdoor path cannot compensate for leaving Z unharvested. For instance, given $\beta = 1$, $X = 1$, $Z = 1$ and $\gamma > 0$, Exhibit 3 shows the underperformance (colored in red) of factor strategies that under-control for a confounder, relative to correctly specified factor strategies. As an example, for $(X, Z, \beta, \gamma, \delta) = (1, 1, 1, 3, 1)$, then $E[YE[\hat{Y}|X, Z]|X, Z] = 16$ while $E[YE[\hat{Y}|X]|X] = 6.25$, even though $E[\hat{\beta}|X] = 2.5$ is estimated with the correct sign ($\beta = 1$). The second and third quadrants contain regions where under-controlling for a confounder causes factor strategies to outperform (colored in blue), however the reason is the misuse of the correctly specified strategy: for $\gamma < 0$, a rational investor would not target securities with $Z = 1$. Those two quadrants turn red for $(X, Z) = (1, -1)$. We merely report results of $\gamma < 0$ for completeness.

3.1.2. FORECASTING STRATEGIES

Consider an investor who wishes to monetize the forecasting power of a correctly specified factor model. In order to do so, he develops a strategy that buys or sells securities based solely on the forecasted returns $E[\hat{Y}|X, Z]$, regardless of that security's factor exposure X . Since the investor is not targeting factor exposures, there is no conditioning on X or Z . A forecasting strategy that sizes positions by $E[\hat{Y}|X, Z]$ produces (unconditional) expected returns

$$\begin{aligned}
E[YE[\hat{Y}|X, Z]] &= E[Y(X\beta + Z\gamma)] \\
&= \beta E[XY] + \gamma E[YZ] \\
&= \beta(\beta(1 + \delta^2) + \delta\gamma) + \gamma(\gamma + \beta\delta) \\
&= \beta^2 + \beta^2\delta^2 + 2\beta\delta\gamma + \gamma^2 \\
&= \beta^2 + (\beta\delta + \gamma)^2 \geq 0
\end{aligned} \tag{8}$$

However, with the incorrect model specification $Y = X\beta + \varepsilon$, the expected returns of a forecasting strategy that sizes positions by $E[\hat{Y}|X]$ are

$$\begin{aligned}
E[YE[\hat{Y}|X]] &= E\left[YX\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right)\right] \\
&= (\beta(1 + \delta^2) + \gamma\delta)\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right) \\
&= (1 + \delta^2)\left(\beta + \frac{\gamma\delta}{1 + \delta^2}\right)^2 \geq 0
\end{aligned} \tag{9}$$

The conclusion from the above equations is that under-controlling for a confounder always causes forecasting strategies to underperform.

Exhibit 4 summarizes the earlier results, which debunk two widespread myths: (a) that specification errors are not dangerous to factor investors as long as correlations remain constant and the factor model estimates β with the correct sign; and (b) that specification errors do not impact the performance of forecasting strategies. In presence of a confounder Z , an under-

controlling model will cause the underperformance of both, factor strategies and forecasting strategies.

Confounder Case	Correct Specification	Under-Controlling
Factor Strategy	$(X\beta + Z\gamma)^2$	$X^2 \left(\beta + \frac{\gamma\delta}{1 + \delta^2} \right)^2$
Forecasting Strategy	$\beta^2 + (\beta\delta + \gamma)^2$	$(1 + \delta^2) \left(\beta + \frac{\gamma\delta}{1 + \delta^2} \right)^2$

Exhibit 4 – Scorecard of performance for factor and forecasting strategies in case of a confounder with constant correlations

Appendix 4 extends this analysis to the case when there is a shift in δ , and demonstrates that this is enough to cause systematic losses in factor and forecasting strategies that fail to control for a confounder, even if the risk premia (β, γ) remain constant.

3.2. CONTROLLING FOR A COLLIDER

Consider a data-generating process with equations:

$$\begin{aligned} Y &= X\beta + u \\ Z &= Y\gamma + X\delta + v \end{aligned} \tag{10}$$

where $\gamma \neq 0$, $\delta \neq 0$, and variables (u, v, X) are independent and identically distributed as a standard Normal, $(u, v, X) \sim N[0, I]$. The causal graph for this process is displayed in Exhibit 5. In the language of causal inference, Z is a collider, because it is influenced by both the treatment (X) and the outcome (Y). The non-causal path from X to Y through Z (i.e., $X \rightarrow Z \leftarrow Y$) is blocked, unless the model controls for collider Z .

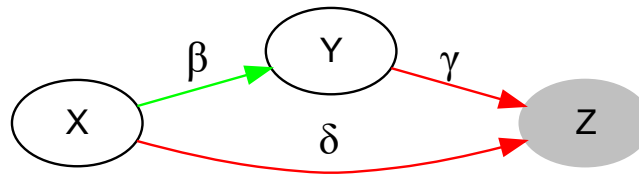


Exhibit 5 – Variable Z as a controlled collider

The data-generating process is unknown to observers. With proper knowledge of the causal graph, observers would estimate the effect of X on Y (green arrow in Exhibit 5) by fitting the equation $Y = X\beta + \varepsilon$. With that model specification, $E[Y|X] = E[\hat{Y}|X]$, i.e. the true value of Y (from the data-generating process) is on average equal to the value of Y expected by observers, and the parameter is estimated without bias, $E[\hat{\beta}|X] = \beta$.

Suppose that observers incorrectly attempt to estimate the causal effect of X on Y by fitting the equation $Y = X\beta + Z\theta + \varepsilon$ on a sample produced by the process. Controlling for collider Z

(shaded node in Exhibit 5) unblocks the non-causal path (red arrows in Exhibit 5). Then, β and θ are estimated with a bias, with expected values,⁹

$$E[\hat{\beta}|X, Z] = \frac{\beta - \delta\gamma}{1 + \gamma^2} \quad (11)$$

$$E[\hat{\theta}|X, Z] = \frac{\gamma}{1 + \gamma^2} \quad (12)$$

See Section A.2 for a proof of the above statements. Parameters (δ, γ) may bias the estimate of risk premium β (note that the bias is zero when $(\delta, \gamma) = (0, 0)$). For appropriate values of (δ, γ) , the bias can be so large that it flips the sign of the estimated β and θ (a phenomenon that biostatisticians call Berkson's fallacy, named after the physician who discovered it, see Berkson [1946]).¹⁰ Accordingly, observers expect that the value of Y is

$$\begin{aligned} E[\hat{Y}|X] &= E[XE[\hat{\beta}|X, Z] + ZE[\hat{\theta}|X, Z]|X] \\ &= XE[\hat{\beta}|X, Z] + E[Z|X]E[\hat{\theta}|X, Z] \\ &= X \frac{\beta - \delta\gamma}{1 + \gamma^2} + X \frac{\gamma}{1 + \gamma^2} (\beta\gamma + \delta) \\ &= X\beta \end{aligned} \quad (13)$$

$$E[\hat{Y}|X, Z] = X \frac{\beta - \delta\gamma}{1 + \gamma^2} + Z \frac{\gamma}{1 + \gamma^2} \quad (14)$$

whereas the true value (from the data-generating process) of Y has expectation

$$E[Y|X] = X\beta \quad (15)$$

Notably, the model that over-controls for a collider estimates Y with lower variance of the error than the correctly specified model, i.e. $Var[\varepsilon] < Var[u]$ (see Section A.3.1 for a proof).

From an investment standpoint, only variable X is a factor exposure, because only X is a cause of Y . The correctly specified model is $E[\hat{Y}|X] = XE[\hat{\beta}|X] = X\beta = E[Y|X]$, thus Y is only influenced by parameter β . Therefore, only β is a risk premium, and (δ, γ) are not risk premia. Next, we study the implications of over-controlling for a collider on the performance of factor strategies and forecasting strategies, when system parameters (β, γ, δ) do not change, hence all correlations remain constant.

⁹ Since θ is absent from the correctly specified model, its unbiased estimate should be $E[\hat{\theta}|X, Z] = 0$.

¹⁰ This bias in the parameter estimation is not in contradiction with the least-squares property whereby $E[Y - E[\hat{Y}|X, Z]] = 0$.

3.2.1. FACTOR STRATEGIES

Consider an investor who wishes to harvest the risk premia offered by securities with returns Y , where the data-generating process is the one described in Section 3.2. With the correctly specified factor model, the factor strategy that sizes positions by $E[\hat{Y}|X]$ produces expected returns

$$\begin{aligned} E[YE[\hat{Y}|X]|X] &= E[Y|X]X\beta \\ &= E[X\beta + u|X]X\beta \\ &= (X\beta)^2 \geq 0 \end{aligned} \tag{16}$$

The above equation shows that the factor strategy based on the correctly specified factor model extracts the only risk premium, β , as designed. The collider parameters (γ, δ) are absent from the equation.

However, with the incorrect model specification $Y = X\beta + Z\theta + \varepsilon$, the investor attempts to design a factor strategy that profits from securities' exposures X and Z , instead of from the only factor exposure, X . One problem with this attempt is that collider Z is a function of Y , and not the other way around. This model specification is an instance of reverse causation. By the time Z is known and it is possible to condition on its value, the value of Y has already been set. In other words, it is not possible to estimate $E[\hat{Y}|X, Z]$ *before* the value of Y has been set. If the investor knew that Z is a function of Y , he could at least condition on $E[\hat{Z}|X] = X(\beta\gamma + \delta)$, but this is inconsistent with the investor's (false) assumption that Z is a cause of Y . Therefore, attempting to condition on a value of Z before the value of Y is set is tantamount to conditioning on a variable $\tilde{Z}$ that adds no information beyond what is known when the value of X is set. Formally, $E[\cdot|X, \tilde{Z}] = E[\cdot|X]$, thus $E[u|X, \tilde{Z}] = E[u|X] = 0$. The expected returns of a strategy that sizes positions by $E[\hat{Y}|X, \tilde{Z}]$ are

$$\begin{aligned} E[YE[\hat{Y}|X, \tilde{Z}]|X, \tilde{Z}] &= E[(X\beta + u)(E[\hat{\beta}|X, Z]X + E[\hat{\theta}|X, Z]\tilde{Z})|X, \tilde{Z}] \\ &= (X\beta + E[u|X, \tilde{Z}])E[\hat{\beta}|X, Z]X + E[\hat{\theta}|X, Z]\tilde{Z} \\ &= (X\beta + E[u|X, \tilde{Z}])\frac{\beta X + \gamma(\tilde{Z} - \delta X)}{1 + \gamma^2} \\ &= X\beta\frac{X\beta + \gamma(\tilde{Z} - \delta X)}{1 + \gamma^2} \end{aligned} \tag{17}$$

This factor strategy has failed to achieve the intended expected return, $(X\beta + Z\theta)^2$, which would have been achieved if Z had been a confounder or an independent cause of Y . Controlling for collider Z has unblocked the non-causal path $X \rightarrow Z \leftarrow Y$, with the effect of creating a non-causal association between X and Y . Accordingly, now the collider parameters (γ, δ) appear in the equation for expected strategy returns, despite of not being risk premia. Even if $\beta = 0$, this over-controlled factor strategy is still exposed to that non-causal association.

In the above equation, we have assumed that $E[u|X, \tilde{Z}] = 0$, however this is not a critical assumption. Regardless of the value of $E[u|X, \tilde{Z}]$, the conclusion holds that for appropriate combinations of real values of $(X, \tilde{Z}, \beta, \gamma, \delta)$, the strategy yields systematic losses, i.e.

$E[YE[\hat{Y}|X, \tilde{Z}]|X, \tilde{Z}] < 0$. For example, the strategy yields systematic losses under $(X, \tilde{Z}, \beta, \gamma, \delta) = (1, 1, 1, 1, 3)$.

Exhibit 6 demonstrates the perverse effect of over-controlling for a collider on the performance of a factor investing strategy. The causal graph shows that factor X (the cause) influences variable Y (the target) with factor loading $\beta = 2$ (the effect), and that variable Z is influenced by variables X and Y , thus Z is a collider. First, consider a researcher who deduces from a causal discovery analysis that Z is a potential collider, and accordingly excludes Z as a control. The green line shows that the factor strategy is equally profitable regardless of the effect size of Y on Z ($\delta = 1$ or $\delta = 3$). The reason is, as proved earlier, a causal model is robust to shifts in the values of non-causal relations. Second, consider a researcher who introduces Z as a control, possibly because doing so maximizes the model's adjusted R-squared. As a result of over-controlling for Z , the expected value of $\hat{\beta}$ is biased. When $\delta = 1$, then $E[\hat{\beta}|X, Z] = 0.5$, and the misspecified strategy still performs well (see red line before the dashed vertical line), because the estimated $\hat{\beta}$ has the correct sign. However, if the value of δ changes from $\delta = 1$ to $\delta = 3$, then $E[\hat{\beta}|X, Z] = -0.5$, and the misspecified strategy yields systematic losses (see red line after the dashed vertical line), because an investor seeking exposure to X will sell securities that should be bought and vice versa. Note that the misspecified strategy yields systematic losses despite the fact that the only true risk premium (β) remains constant. López de Prado and Zoonekynd [2025] show numerous examples of this phenomenon in widely used Barra models.

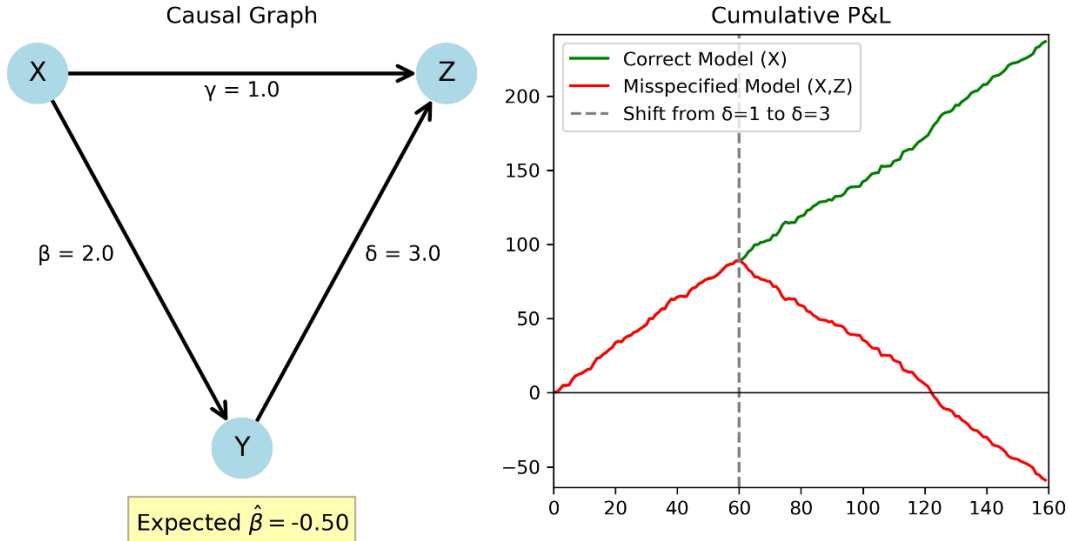


Exhibit 6 – Systematic losses caused by a collider

3.2.2. FORECASTING STRATEGIES

With the correctly specified factor model, a forecasting strategy that sizes positions by $E[\hat{Y}|X]$ has expected returns

$$\begin{aligned}
 E[YE[\hat{Y}|X]] &= E[(X\beta + u)X\beta] \\
 &= \beta^2 E[X^2] + \beta E[Xu] \\
 &= \beta^2 \geq 0
 \end{aligned} \tag{18}$$

Let us turn our attention to a model that over-controls for a collider Z . As argued earlier, attempting to condition on a value of Z before the value of Y is set is tantamount to conditioning on a variable $\tilde{Z}$ that adds no information beyond what is known when the value of X is set. For instance, this would be the case if $\tilde{Z}$ represents the previous value of Z , in which case $E[Y\tilde{Z}] = 0$. Then, the forecasting strategy that sizes positions by $E[\hat{Y}|X, \tilde{Z}]$, has expected returns

$$\begin{aligned}
E[YE[\hat{Y}|X, \tilde{Z}]] &= E[(X\beta + u)(E[\hat{\beta}|X, Z]X + E[\hat{\theta}|X, Z]\tilde{Z})] \\
&= E[(X\beta + u)E[\hat{\beta}|X, Z]X] + E[(X\beta + u)E[\hat{\theta}|X, Z]\tilde{Z}] \\
&= \beta E[\hat{\beta}|X, Z]E[X^2] + E[\hat{\beta}|X, Z]E[Xu] + E[\hat{\theta}|X, Z]E[Y\tilde{Z}] \quad (19) \\
&= \beta \frac{\beta - \gamma\delta}{1 + \gamma^2} + \frac{\gamma}{1 + \gamma^2} E[Y\tilde{Z}] \\
&= \beta \frac{\beta - \gamma\delta}{1 + \gamma^2}
\end{aligned}$$

In the above equation, we have assumed that $E[Y\tilde{Z}] = 0$, however this is not a critical assumption. Regardless of the value of $E[Y\tilde{Z}]$, the conclusion holds that for appropriate combinations of real values of (β, γ, δ) , the strategy yields systematic losses, i.e. $E[YE[\hat{Y}|X, \tilde{Z}]] < 0$. For example, the strategy yields systematic losses under $(\beta, \gamma, \delta) = (1, 1, 2)$.

Exhibit 7 summarizes the above results. They evidence that over-controlling for a collider opens the possibility for systematic losses, even if correlations remain constant and the risk premium is estimated with the correct sign. In contrast, the downside of under-controlling for a confounder was limited to underperforming the strategy based on the correctly specified model, without the possibility of systematic losses (see Exhibit 4). The implication is that over-controlling for a collider is more dangerous to investors than under-controlling for a confounder.

Collider Case	Correct Specification	Over-Controlling
Factor Strategy	$(X\beta)^2$	$X\beta \frac{X\beta + \gamma(\tilde{Z} - \delta X)}{1 + \gamma^2}$
Forecasting Strategy	β^2	$\beta \frac{\beta - \gamma\delta}{1 + \gamma^2}$

Exhibit 7 – Scorecard of performance for factor and forecasting strategies in case of a collider with constant correlations

Appendix 5 extends this analysis to the case when there is a shift in γ or δ , and demonstrates that this is enough to cause systematic losses in factor and forecasting strategies that control for a collider, even if the risk premium β remains constant.

3.3. ROBUSTNESS AS A VIRTUE OF CAUSALITY

The results derived in Sections 3.1 and 3.2 allow us to draw three important conclusions. First, even if all correlations remain constant, misspecified factor and forecasting strategies: (a) underperform the correctly specified strategy in the confounder case; and (b) can yield systematic

losses in the collider case. This is true regardless of whether the risk premia were estimated with the correct sign.

Second, the performance of factor strategies that under-control for a confounder is not robust to parameter shift, and these strategies can yield systematic losses, even if the risk premia remain constant and are estimated with the correct sign.

Third, the performance of correctly specified factor and forecasting strategies is robust to parameter shifts. The performance of correctly specified factor strategies is unaffected by changes in both, the confounding effect and the collider parameters. The performance of correctly specified forecasting strategies is unaffected by changes in the collider parameters, but it is impacted (positively or negatively) by changes in the confounding effect, although not to the point of yielding systematic losses. Accordingly, robustness is neither a virtue of a model's complexity nor simplicity, but of the causal content injected by the modeler.

Several authors have argued that the erratic performance of factor investing strategies is due to time-varying risk premia (see, for example, Evans [1994], Anderson [2011], Cochrane [2011]). Our findings show that this erratic performance is better explained (in simpler, more parsimonious terms) by specification errors, without requiring changes in the market's compensation for risk, or changes in investor behavior. The risk premia can remain constant, and specification errors will still cause factor investing strategies to perform erratically.

Factor strategies often involve only a handful of parameters, but even then the risk of over-controlling for a collider is high. The next sections analyze how the econometric canon favors over-controlled models, which in turn explains the underperformance and systematic losses experienced by factor investors.

4. EXPLANATORY POWER AND P-VALUES UNDER OVER-CONTROLLING

In statistics, the explanatory power of a model indicates how well that model fits a set of observations. By far the most common way to express a factor model's explanatory power is through its R-squared (also known as its coefficient of determination), or its adjustment for the number of variables, known as the adjusted R-squared.

Section A.3 contains proofs to the following statements: (a) the R-squared of a factor model that over-controls for a collider always exceeds the R-squared of the correct model; (b) given enough observations, the adjusted R-squared of a factor model that over-controls for a collider exceeds the adjusted R-squared of the correct model; and (c) when the collider effects are strong and of similar magnitude, or the causal effect is weak, the p -values of estimated coefficients in a factor model that over-controls for a collider are smaller than those of the correct model.

In the case of the adjusted R-squared, the number of observations (T) needed for choosing the over-controlled model is $|\gamma| > \frac{1}{\sqrt{T-3}}$, where $|\gamma|$ is a function of the strength of the collider (relative to the standard Gaussian noise). For instance, for the value of γ used in Section 3.2.1's example ($\gamma = 1$), the over-controlled model has a higher adjusted R-squared than the correct model when $T > 4$. Like the adjusted R-squared, alternative associational methods designed to penalize model

complexity fail to address this problem, because only causal approaches can target the specific issue of controlling for a collider.

In the case of p -values, false positives are more likely when the collider effects in an over-controlled model are sufficiently strong and of similar magnitude ($|\delta| \gg 0$, $|\gamma| \gg 0$, $|\delta| \approx |\gamma|$) or when the causal effect is weak ($|\beta| \approx 0$). The implication is that the presence of colliders facilitates p -hacking under rather general conditions.

5. SPECIFICATION SEARCHING

The econometric canon, as taught in the most widely accepted textbooks, advocates choosing the model specification that yields the highest explanatory power. This conflates two separate steps of the modeling process: model identification and model estimation (López de Prado [2023a, section 6.4.2.3]). For example, Greene [2012, section 7.4] favors the use of R-squared for choosing among competing specifications when the model is intended for within-sample prediction (as it is common in factor investing papers), and favors the use of adjusted R-squared for choosing among specifications when the model is intended for forecasting.

For the reasons presented in Section 4, this approach to specification selection is particularly misleading in the presence of colliders. However, the econometric canon is seemingly ignorant of the problem of collider bias, discussed in Section 3.2. For instance, Greene [2012, p. 136] indicates that the downside of including irrelevant variables is “the reduced precision of the estimates,” and that “[o]mitting variables from the equation seems generally to be the worse of the two errors.” Furthermore, Greene [2012, p. 137] recommends a “general-to-simple” approach, where a researcher starts with a “kitchen sink regression, which contains every variable that might conceivably be relevant.” Those relevant variables certainly include every possible collider, all but guaranteeing that the selected model is misspecified.

This state of confusion is not particular to Greene [2012], but a misunderstanding prevalent throughout the econometrics literature. The root problem is that econometricians tend to choose their specifications through associational arguments (such as R-squared), rather than through the discovery and analysis of the causal structure of the data-generating process. In the words of Leamer [1978], “[w]hen a specification is chosen to fit the data well, the resulting inference is conditional on that choice, and classical statistical measures are no longer valid.” Spirtes et al. [2000, chapter 8] offer the following warning: “In the absence of very strong prior causal knowledge, multiple regression should not be used to select the variables that influence an outcome or criterion variable in data from uncontrolled studies.” Chen and Pearl [2013] reviewed six of the most popular textbooks in econometrics, concluding that they “deviate significantly from modern standards of causal analysis”.¹¹

These flawed specification selection procedures have been applied to factor investing, as epitomized by some of the most influential publications in this field. Fama and French [2015] argument for adding two factors to their initial three-factor model specification was that “the five-

¹¹ There are remarkable exceptions to econometrics’ state of causal confusion. In the year 2021, Guido Imbens and Joshua Angrist received (in conjunction with David Card) the Nobel Memorial Prize in Economics in recognition “for their methodological contributions to the analysis of causal relationships.” These two authors have fought for decades to instill the rigor of causal inference into econometrics, or as Leamer [1983] put it, “take the con out of *econometrics*.”

factor model performs better than the three-factor model when used to explain average returns.” Similarly, the three-pass factor regression approach advocated by Giglio and Xiu [2021] applies principal components analysis (PCA) to the matrix of asset returns to extract latent variables—i.e., directions of maximal variance in returns unexplained by observed factors—, assuming without causal evidence that those latent variables must be confounders. Unfortunately, as proven in Section 4, if any of these variables is a collider, the model’s estimates are biased, with the potentially dire consequences for investors discussed in Section 3.

Mounting empirical evidence points in that direction. Sadeghi et al. [2024] apply the PC algorithm for causal discovery to the variables used in Fama and French [1993], and find that the HML factor may be a collider (see figure 8b in that publication). In addition, Gu et al. [2023] present empirical evidence that the factor model regression in Fama and French [2015] is also misspecified, with the presence of at least one collider (market returns net of the risk-free rate).

Kelleher and Mittal [2025] analyze factor investing using a causal inference framework, highlighting how common econometric practices introduce confounder and collider biases. They show that cost surprises initially appear positively related to returns because sales volume—an unaccounted confounder—increases both sales and costs, creating a spurious association. Once properly controlling for sales surprises, the true negative impact of cost surprises on returns is revealed. This example illustrates how failing to adjust for confounders can lead to incorrect conclusions about investment signals. Conversely, they also warn against inadvertently introducing collider bias. For instance, conditioning on sales surprises—when both cost surprises and returns influence it—can open a non-causal path that distorts inference. Their study demonstrates that relying solely on explanatory power, without understanding the causal structure, risks misspecification and financial losses.

While factor investing has traditionally been presented as an empirical, data-driven approach, its logic is fundamentally causal: the econometric structure of factor models implies that certain risk characteristics cause differences in expected returns (see López de Prado [2023a, 2023b]). Investors are not merely interested in associating factors with returns, but in isolating and monetizing the true causal effects of these exposures. Misidentifying these drivers, or incorrectly estimating their premia due to specification errors, undermines the strategy’s validity and profitability. Furthermore, López de Prado et al. [2025] show that factor portfolios devoid of a causal theory are likely inefficient.

5.1. FINDING COLLIDERS IN BARRA FACTORS

To demonstrate these ideas, we implement the PC causal discovery algorithm introduced by Spirtes et al. [2000] on the daily return series of risk factors from 85 Barra risk models. Exhibit 8 summarizes the resulting causal structures by keeping only those edges that appear in at least one-third of the individual graphs.

Exhibit 9 presents the corresponding adjacency matrix. The two entries below the main diagonal highlight edges where the PC algorithm could not resolve the direction of causality, specifically between (“size”, “liquidity”) and (“size”, “resvol”). Using this inferred graph, an investor can establish an appropriate model specification. Specifically, when targeting a given factor, the investor should condition on that factor’s parents rather than its descendants, thus preventing bias

from conditioning on a collider. For example, when investing in “growth,” one should adjust for “momentum,” “leverage,” “long-term reversal,” “profit,” and “investor quality,” while avoiding controls for value-related factors such as “earnings yield,” “book-to-price,” “earnings quality,” and “dividend yield.” Exhibit 10 marks suitable controls in green, those to avoid in red, and irrelevant variables in grey.

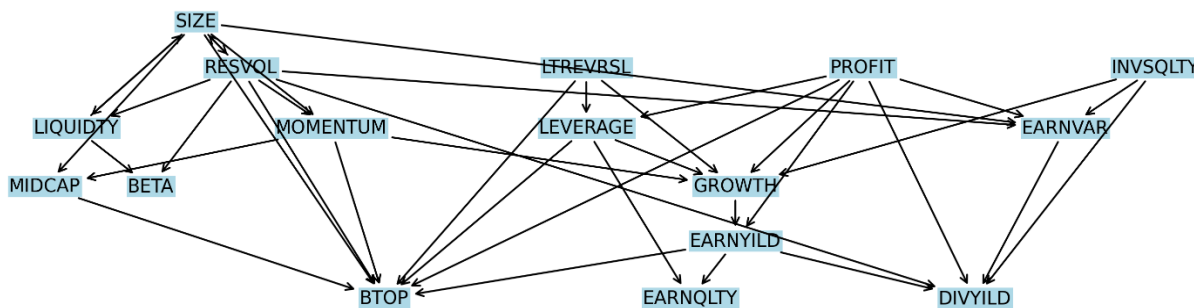


Exhibit 8 – Combined causal structure inferred using the PC algorithm

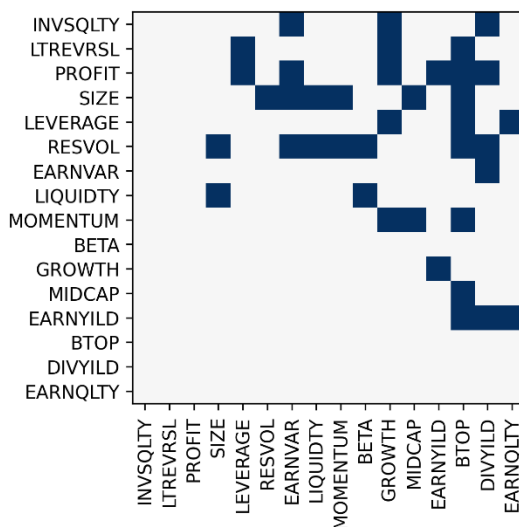


Exhibit 9 – Adjacency matrix

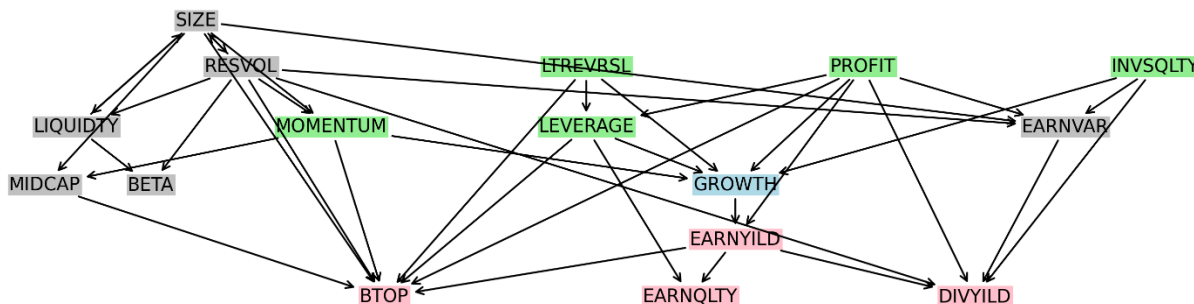


Exhibit 10 – Valid (green) and invalid (red) controls for a Growth factor model

Consider one of these risk models, for example USE4L. We attempt to predict “growth” using the other factors, selecting predictors to maximize the adjusted R-squared. In Exhibit 11, the left panel

displays the adjusted R-squared values for models that include all descendants, with a greedy algorithm sequentially adding parent factors (shown on the y-axis). The complete model — incorporating all descendants and all parents from the y-axis — achieves an adjusted R-squared of about 8.5%. However, these models are misspecified because descendants should not be included as controls. The right panel of Exhibit 11 illustrates models that exclude descendants, again using a greedy approach to add parent factors step by step. The final (properly specified) model in this case, which uses all parents but no descendants, yields an adjusted R-squared around 7.8%, appearing “worse” than the incorrectly specified model.

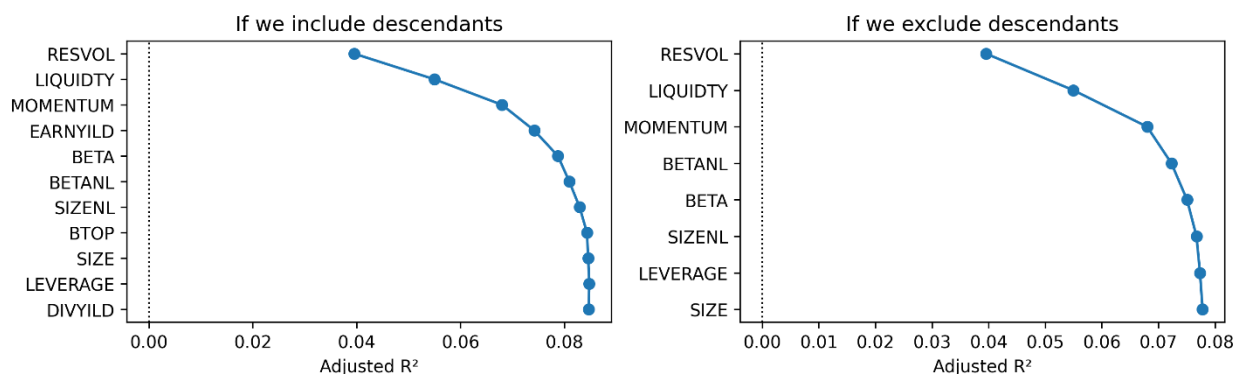


Exhibit 11 – Adjusted R-squared for misspecified (left) and properly specified (right) factor models

Our findings stand in contrast to recent claims, often framed in terms of “double descent,” suggesting that greater model complexity can improve generalization performance. As Cartea et al. [2025] show, this narrative does not hold in noisy financial environments, where increased complexity amplifies estimation error and leads to deteriorating portfolio performance, despite superficially stronger in-sample metrics. Their findings support our notion of the factor mirage: an illusion of model quality created by canonical econometric practices that reward overfitting under the guise of rigor. The implication is that blindly adding controls or complexity—while compliant with the econometric canon—may increase the likelihood of systematic investor losses.

The burden of proof is on factor researchers promoting an investment opportunity, to present theoretical and empirical evidence that the chosen variables do not include potential colliders. Absent that evidence, there is a significant risk that the model is over-controlled, and that investors are exposed to adverse outcomes.

6. THE CAUSAL MECHANISM FOR ADVERSE OUTCOMES

Virtually all factor researchers apply econometric procedures that fundamentally date back to the 1930s (when the Econometric Society and the Cowles commission were launched) and even the late 18th century (when Gauss developed the least squares method, see Stigler [1981]), before the development of causal inference. For the reasons explained earlier, the current econometric canon misleads researchers into selecting a certain class of misspecified factor models, in place of correctly specified models. Factor researchers (who selected the misspecified factor models) build investment portfolios that are more likely to underperform and yield systematic losses, to the benefit of more informed investors (who applied causal discovery to select correctly specified

models). Consequently, factor investors and informed investors tend to take opposite sides of the same trade.

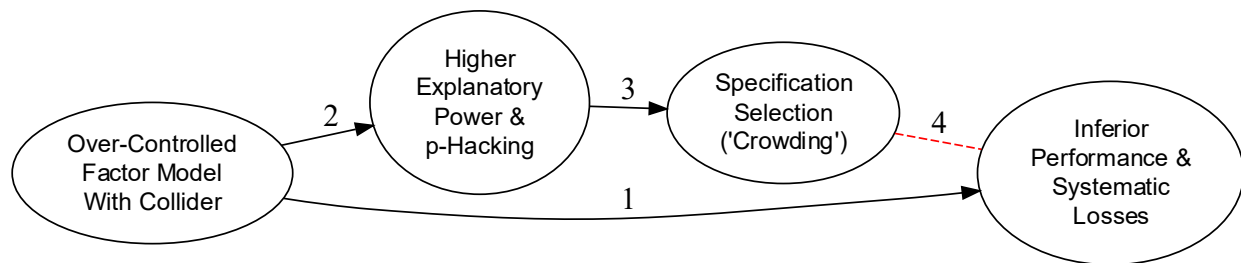


Exhibit 12 – The causal mechanism behind the failure of factor investing

Exhibit 12 outlines the causal mechanism that enables adverse outcomes in factor investing. In Section 3, we proved that factor and forecasting strategies based on factor models that over-control for a collider can yield systematic losses, even if correlations remain constant and the risk premium was estimated with the correct sign (link 1 in the causal mechanism). In Section 4, we proved that misspecified factor models that over-control for a collider exhibit a higher R-squared than correctly specified models. We also proved that, under general conditions, colliders make it easier to *p*-hack a factor (link 2 in the causal mechanism). In Section 5, we explained how the canonical approaches for specification-selection in econometrics favor over-controlled models, crowding out correctly specified models (link 3 in the causal mechanism). The combined effect is an increased association between selected (either published or deployed) factor investing strategies and underperformance, including systematic losses (link 4 in the causal mechanism).

There are two reasonable counter-arguments to Exhibit 12. First, factor investing papers usually publish long historical backtests with a positive expected return. Could those positive backtests indicate that those strategies' models are not over-controlled? The answer is no. As the False Strategy Theorem has proven (Bailey et al. [2014], López de Prado and Bailey [2021]), it is trivial to overfit a backtest to simulate any desired performance. For example, a researcher can run thousands of backtests by trying alternative investment universes and time periods, introducing arbitrary profit-taking and stop-out rules, or imposing *ad hoc* constraints to the portfolio optimization, to cite a few common variations. The expected Sharpe ratio of the best backtest rises with the number of trials, even if the true Sharpe ratio is zero. Historical backtests do not model the data-generating process, hence they cannot tell us much about whether the factor model is correctly specified. A better tool for assessing specification errors is causal discovery algorithms. A second counter-argument to Exhibit 12 is that some popular factor investing strategies have delivered positive out-of-sample performance for multiple years, before experiencing losses or flattening out. Could this be evidence that these strategies' models are not over-controlled? Again, the answer is no. The sheer number of factors, many of which are noisy (Akey et al. [2023]), has resulted in thousands of alternative implementations. Some of those implementations will perform well out-of-sample for a while (a statistical fluke). Also, a strategy's popularity can push up the prices of securities with targeted characteristics, driving up the strategy's live performance for several years (a self-fulfilling prophecy), before losses offset the initial trend. The factor investing industry continues to grow at sufficient pace to potentially mask its true expected returns for years.

Cochrane [2011] famously described the explosion of factor claims as the “factor zoo,” which has been attributed to widespread p -hacking and multiple testing (e.g., Bailey et al. [2014], Harvey et al. [2016]). While brute-force data mining remains a plausible explanation, our findings suggest a deeper and more structural mechanism: the factor mirage. A factor mirage occurs when canonical econometric practices systematically favor over-controlled, misspecified models that produce higher R-squared values and lower p -values than the correctly specified, genuinely explanatory model. This bias—often due to collider structures—can lead to p -hacking outcomes after just a few specification trials, even in the absence of ill intent. In this sense, the econometric canon enables what we term few-shot p -hacking, where researchers inadvertently produce misleading yet statistically compelling results by following accepted procedures. The factor mirage, then, is not just a product of undisciplined testing, but a structural illusion embedded in standard modeling conventions.

7. OUTLINE OF A RESEARCH PROTOCOL FOR CAUSAL FACTOR INVESTING

The causal mechanism for adverse outcomes is enforced by canon-complying factor researchers (a self-inflicted wound), not by the informed investors taking the other side of the trades. The solution is to correct the econometric canon such that the causal mechanism is disrupted. This section provides a practical solution for practitioners and academics.

An analysis of Exhibit 12 suggests two primary corrections to the econometric canon. First, utilize the tools of causal inference. The correct way to justify a factor model’s specification is to discover the causal graph associated with the data-generating process, and to apply do-calculus to block all non-causal paths, thus avoiding both, under-controlling for a confounder and over-controlling for a collider. Second, prevent that over-controlled models crowd out correctly specified factor models. Academics and practitioners should desist in their practice of choosing the factor model’s specification with the highest explanatory power, as this practice all but guarantees over-controlling, with potentially dire consequences for investors. Investors can also take matters into their own hands, by defunding investment strategies that do not have a solid causal foundation (see López de Prado [2015, 2023a]).

Next, we outline a new econometric protocol for the discovery of factors, which complies with modern machine learning and causal inference techniques. This protocol ensures correct model specification, avoiding both confounder and collider biases, while preserving predictive performance. It is divided into the following seven steps: (1) Variable selection; (2) causal discovery; (3) causal adjustment set; (4) causal explanatory and predictive power; (5) causal portfolio construction; (6) backtest; and (7) multiple testing adjustments.

Step 1: Variable Selection

The researcher identifies the potential variables associated with contemporary or future returns, depending on the purpose of the factor model (attribution or risk premia harvesting). Potential variables should be identified using flexible machine learning tools that capture complex relationships without assuming a specific model form. Three types of tools can help in this task: (1.a) non-parametric methods for detecting mutual information between factors and returns; (1.b) in-sample analysis of the influence of various predictors on the returns, computing Shapley values, mean decrease impurity (MDI), etc.; (1.c) cross-validation feature importance methods, such as permutation feature importance (PFI). To prevent leaking test-set information into the train-set, it

is important to ensure that returns do not overlap (i.e., the time window used to compute a return ends where the next return's time window begins). If returns do overlap, researchers should apply a purged cross-validation algorithm. Also, if there is a strong time series dependence, it is necessary to apply an embargo to some train-set observations immediately after the end of the test-set. See López de Prado [2018, chapter 7] for details on purged cross-validation algorithms.

Step 2: Causal discovery

The researcher estimates the causal graph that characterizes the structure of dependencies between all variables identified in (1.a), (1.b) and (1.c), through the application of causal discovery algorithms (see Glymour et al. [2019]), economic reasoning, and domain knowledge. The causal graph may reflect: (2.a) investor ex-ante views (Laudy et al. [2022]); or (2.b) observed outcomes (Olivetti et al. [2025], Sadeghi et al. [2024], Gu et al. [2023]). Causal discovery algorithms (e.g., PC, LiNGAM), can be valuable even if they are not exact, as they help eliminate graphs inconsistent with empirical evidence. From this refined set, researchers can explain which graph is most consistent with economic intuition, making explicit assumptions that would otherwise remain hidden. While the chosen graph may still be incorrect, at least the researcher has disclosed their choices, subjecting them to peer-review and refutability (López de Prado [2023a]).

Step 3: Causal adjustment set

The researcher applies do-calculus on (2.a) or (2.b) to propose the set of controls that close all backdoor paths, while leaving only open the causal path (Pearl [2009], Pearl et al. [2016]). In particular, the researcher will attempt to control for all confounders in the causal graph, while avoiding mediators and colliders. Special types of control methods include: (3.a) backdoor adjustment; (3.b) front-door adjustment; and (3.c) instrumental variables. Each control variable must be economically and causally justified to prevent mechanical overfitting.

Step 4: Causal Explanatory and Predictive Power

The researcher presents empirical evidence of the explanatory or predictive power of the causal model, in the form of an estimated generalization error. This analysis could cover complementary aspects: (4.a) the probability of discretized returns (a binary or multi-class classification problem); (4.b) the ranking of discretized returns (a binary or multi-class classification problem); (4.c) the size of returns (a regression problem). Models that perform well in one of these aspects but not the rest could still be turned into profitable strategies, and this analysis will inform the strategy's design. Evidence for (4.a) can be presented in the form of purged cross-validated log loss or Brier scores, which help assess how well the model has calibrated probabilities of potential outcomes. While both scoring methods penalize incorrect predictions made with high confidence, log loss tends to be more helpful at identifying and managing tail risks, and Brier tends to be more helpful with overall performance. An instance of (4.a) of particular relevance is the explanation or prediction of the probability of a positive or negative return (a binary classification problem). Evidence for (4.b) can be presented in the form of purged cross-validated receiver operating characteristic (ROC) curves, precision-recall (PR) curves, mean reciprocal ranking, or accuracy. These scores help assess how well the model has calibrated the ranking of potential outcomes. Evidence for (4.c) can be presented in the form of purged cross-validated mean-square errors, R-squares, Spearman correlation, etc.

Binary classification scores can be applied to multi-class problems (4.a) and (4.b) through One-versus-Rest (OvR) encoding and averaging (micro, macro, weighted, samples, etc.). Macro or sample averaging is generally preferable when some classes are more important (e.g., extreme negative returns, for downside protection), whereas weighted or micro averaging is generally preferable when overall performance is more important.

Each of the above scores can be assessed against the value of a naïve algorithm. For example, in a classification problem, a good model should achieve higher scores than those obtained by an algorithm that predicts classes with the same frequency as observed in-sample (thus ignoring features values). Similarly, in a regression problem, a good model should achieve higher scores than those obtained by an algorithm that predicts values with the same average as observed in-sample.

Purged cross-validation ensures results are robust to temporal dependencies. Importantly, explanatory power (for causal attribution) should be evaluated separately from predictive power (for risk premia harvesting), recognizing their distinct roles.

Step 5: Causal portfolio construction

The researcher uses the causal forecasts or causal attributions from the model to construct a portfolio that reflects the targeted exposure to causal factors while controlling for unintended exposures. Unlike traditional portfolio construction based on associational models, this step emphasizes: (5.a) position sizing based on causal effects, where capital is allocated in proportion to the strength of the causal effect (e.g., expected return or risk premium derived from the causal adjustment set), avoiding weights based on spurious associations (e.g., coefficients from misspecified models); (5.b) risk control through causal graphs, ensuring neutrality towards variables not in the causal path (e.g., colliders or irrelevant variables), or hedging undesired causal exposures. The causal graph helps isolate risk exposures and avoid contaminating the strategy with non-causal influences; (5.c) alignment with economic rationale, whereby investment positions should reflect structural relationships (e.g., long high book-to-price if causally linked to returns), avoiding reverse causation or dependence on coincidental patterns; (5.d) stress testing for causal fragility, by simulating perturbations in the causal graph or parameter values to assess robustness, testing whether small changes to assumptions lead to major changes in portfolio behavior (sensitivity to misspecification); (5.e) portfolio constraints and transaction costs, by implementing realistic constraints (e.g., leverage, turnover limits) and incorporating cost-aware optimization that preserves causal integrity (e.g., avoid rebalancing purely based on noise); (5.f) assessing the impact of constraints and trading rules, by computing the transfer coefficient between the causally-optimal portfolio and the implemented portfolio.

López de Prado et al. [2025] demonstrate the dire consequences of factor model misspecification in the context of portfolio optimization. They show that it is not possible to estimate the efficient frontier without a causal factor model, and that the prevailing (associational) factor modeling paradigm leads to investment inefficiency. Furthermore, they show that the biases introduced by factor model misspecification can be so large as producing portfolios where investors buy what they should sell and vice versa, even if investors had perfect knowledge of the mean and covariances.

Step 6: Backtest

The researcher simulates the performance of a systematic investment strategy that applies the forecasts produced by the model, with the goal of assessing the expected risk and return going forward. There are three fundamental types of backtests: (6.a) walk-forward; (6.b) resampling; and (6.c) Monte Carlo.

Backtest (6.a) assesses the performance of an investment algorithm under the assumption that history repeats itself exactly. A first caveat of (6.a) is that past time series merely reflect one possible path produced by the data-generating process. Accordingly, (6.a) is more likely to yield a descriptive (or anecdotal) than an inferential statement (see López de Prado [2018], chapter 11). A second caveat of (6.a) is the inefficient use of data, as the calculation of the first returns can consume a significant portion of the overall sample (burn-in period). A third caveat of (6.a) is that the data-generating process is never stated: should the process change, the researcher will not be able to decommission the algorithm before it loses money, because they never understood the conditions that made the algorithm work.

Backtest (6.b) addresses the first and second caveats of (6.a), by assessing the performance of an investment algorithm under the assumption that future paths can be simulated through the resampling of past observations. The resampling can be deterministic (e.g., jackknife, cross-validation) or random (e.g., subsampling, bootstrap). Because (6.b) can produce many different paths, where the historical is just one possibility, it allows researchers to consider more general scenarios consistent with the data-generating process. For instance, through combinatorial purged cross-validation (CPCV), a researcher can bootstrap the distribution of the algorithm's Sharpe ratio, which is harder to overfit and more informative than the single-path Sharpe ratio derived by (6.a). Still, resampling on a finite historical sample may not yield paths representative of the future (see López de Prado [2018], chapter 12).

Backtest (6.c) addresses the three caveats of (6.a), by assessing the performance of an investment algorithm under the assumption that future paths can be simulated via Monte Carlo. This method requires a deeper knowledge of the data-generating process, which may be derived from the statistical analysis of the observations or theory (e.g., market microstructure, institutional processes, economic links, etc.). For instance, economic theory may suggest that two variables are cointegrated, and empirical studies may indicate the range of values that characterize the cointegration vector. Accordingly, researchers can simulate thousands of years of data, where the cointegration vector takes many different values within the estimated range. This is a richer analysis than merely resampling observations from a finite (and likely unrepresentative) set of observations (see López de Prado [2018], chapter 13). Also, see Joubert et al. [2024].

Step 7: Multiple Testing Adjustments

The researcher accounts for the increased probability of a false positive that inevitably results from conducting multiple tests. These adjustments take two main forms: (7.a) p -value corrections; and (7.b) Sharpe ratio corrections. In the case of (7.a), the researcher may apply well-known procedures such as Holm [1979], Hochberg [1988], or Benjamini and Hochberg [1995]. In the case of (7.b), the researcher may report the Deflated Sharpe Ratio, after adjusting for the variances of the backtested Sharpe ratios and the effective number of trials. It is important to note that backtests

are not independent, thus the effective number of trials is less than the total number of backtests. See López de Prado [2020] for details.

Step	Description	Key Methods / Considerations	Purpose / Outcome
Variable Selection	Identify potential explanatory or predictive variables	Shapley values, mutual information, MDI (in-sample); PFI cross-validation (out-of-sample); Purged CV and embargo	Define candidate variables while controlling for data leakage and overfitting
Causal Discovery	Infer the causal structure of dependencies	Causal discovery algorithms (PC algorithm, LiNGAM, etc.); economic reasoning; ex-ante views; peer-reviewed assumptions; domain expertise	Build an explicit causal graph to guide coherent factor modeling
Causal Adjustment Set	Determine the correct set of controls using do-calculus	Backdoor, front-door adjustment; instrumental variables; economic justification of controls	Block backdoor paths, avoid over-controlling, ensure causal validity
Causal Explanatory and Predictive Power	Evaluate model performance through generalization error	Probability calibration for discretized returns (log loss, Brier); ranking calibration for discretized returns (ROC, PR curves, MRR, classification accuracy); regression (MSE, R^2 , Spearman); OvR encoding and averaging	Assess causal explanatory strength and predictive power independently
Causal Portfolio Construction	Construct a portfolio based on causal forecasts or attributions	Ensuring exposure to causal factors and neutrality to non-causal influences; position sizing by causal effects; risk control via causal graphs; economic rationale alignment; transaction cost-aware optimization; stress testing for causal fragility; compute transfer coefficient against causally-optimal portfolio	Implement a robust, causally valid portfolio that reflects true economic relationships and avoids spurious exposures
Backtest	Investment performance simulations	Walk-forward backtest; resampling backtest; combinatorially-purged cross-validation (CPCV); Monte Carlo backtest	Estimate the performance of an investment strategy that uses the model's predictions
Multiple Testing Adjustments	Correct for false discoveries due to multiple testing	p -value adjustments (Holm-Bonferroni, Hochberg, Benjamini-Hochberg); Deflated Sharpe Ratio	Control for multiple testing bias and avoid spurious factor selection

Exhibit 13 – A Research Protocol for Causal Factor Investing

Exhibit 13 summarizes these seven steps. This protocol shifts the focus of factor discovery from spurious correlations and backtest overfitting to scientifically-grounded causal relationships, thus severing the mechanism outlined in Exhibit 12. By integrating machine learning with causal inference, the framework mitigates overfitting risks and enhances economic interpretability,

ensuring factors possess both explanatory and predictive merit. Ultimately, it empowers investors to demand causal validity and defund strategies built on statistical artifacts.

8. CONCLUSIONS

A common belief among researchers and investors is that specification errors are not dangerous, because an investment strategy based on a misspecified factor model will still perform well as long as the risk premia remain constant and are estimated with the correct sign. In this paper, we have proved that this belief is false, for two reasons. First, even if all correlations remain constant, misspecified factor and forecasting strategies: (a) underperform the correctly specified strategy in the confounder case; and (b) can yield systematic losses in the collider case. This is true regardless of whether the risk premia were estimated with the correct sign. Second, the performance of factor strategies that under-control for a confounder is not robust to parameter shift, and these strategies can yield systematic losses, even if the risk premia remain constant and are estimated with the correct sign.

Robustness is a key virtue of (correctly specified) causal models that associational (non-causal) models lack. Causal models are robust to parameter shift, in the sense that changes to parameters outside the causal path have no effect on the performance of those models. This gives a competitive advantage to causal models over associational models in finance. These are not arguments against machine learning or econometrics models in finance, but in favor of a better (causal) machine learning and econometrics models, that combine interdisciplinary research from statistics, economic theory, and experiment design.

The implication is that factor researchers should cease to dismiss as inconsequential concerns regarding specification errors. Authors should justify carefully their specification choices, discuss potential confounders, and argue why the chosen controls cannot be colliders. Put bluntly, without some knowledge of the causal graph, it is possible (even likely) that an investment strategy will be fatally flawed.

Causal discovery and causal inference methods are being applied successfully to untangle the causal relationships in complex systems across various scientific fields (for example, see Runge et al. [2019], Shen et al. [2011]). In contrast, virtually no factor investing studies motivate their specification choices in terms of causal graphs derived from data; or develop refutable economic theories consistent with discovered causal graphs; or provide an analysis of potential colliders, confounders or mediators; or apply do-calculus to derive a sufficient adjustment set. The consequence is that factor researchers routinely make avoidable specification errors. A misspecified model that over-controls for a collider exhibits a higher explanatory power than a correctly specified model. Given the standard econometric practice of choosing a model's specification based on its explanatory power, it is natural for over-controlled models to crowd-out correctly specified models, making it more likely for underperforming and money-losing models to be selected.

Our findings challenge the scientific soundness and long-term profitability of the current multi-trillion-dollar (associational, non-causal) factor investing industry. To overcome these pitfalls, we have proposed adjustments to the prevailing econometric canon, aligning it with recent advances in machine learning and causal inference.

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There are no interests to declare

APPENDIX

A.1. PROOF OF PROPOSITIONS IN SECTION 3.1

Consider a data-generating process with equations:

$$\begin{aligned} X &:= Z\delta + v \\ Y &:= X\beta + Z\gamma + u \end{aligned} \tag{20}$$

where $\gamma \neq 0$, $\delta \neq 0$, and variables (u, v, Z) are independent and identically distributed as a Gaussian. The process is unknown to observers, who attempt to estimate the causal effect of X on Y by fitting the equation $Y = X\beta + \varepsilon$ on a sample produced by the process. Then, the expression $E[\hat{\beta}|X]$ is,

$$E[\hat{\beta}|X] = (X'X)^{-1}X'E[Y|X] \tag{21}$$

First, note that the equation for Y can be rewritten as

$$Y := (Z\delta + v)\beta + Z\gamma + u = Z(\delta\beta + \gamma) + v\beta + u \tag{22}$$

The model can be written

$$\begin{bmatrix} Z \\ v \\ u \\ X \\ Y \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \delta & 1 & 0 \\ \gamma + \beta\delta & \beta & 1 \end{bmatrix} \begin{bmatrix} Z \\ v \\ u \end{bmatrix} = A \begin{bmatrix} Z \\ v \\ u \end{bmatrix} \tag{23}$$

where $(Z, v, u)' \sim N(0, I)$. The variance is then

$$\text{Var} \begin{bmatrix} Z \\ v \\ u \\ X \\ Y \end{bmatrix} = AA' = \begin{bmatrix} 1 & 0 & 0 & \delta & \gamma + \beta\delta \\ 0 & 1 & 0 & 1 & \beta \\ 0 & 0 & 1 & 0 & 1 \\ \delta & 1 & 0 & \delta^2 + 1 & \beta(1 + \delta^2) + \delta\gamma \\ \gamma + \beta\delta & \beta & 1 & \beta(1 + \delta^2) + \gamma\delta & 1 + \beta^2 + (\gamma + \beta\delta)^2 \end{bmatrix} \tag{24}$$

We can read the covariances between X, Y, Z, u, v from this matrix. We can use it to compute conditional expectations (Eaton [1983, pp. 116-117]), for instance:

$$\begin{aligned} E[v|X] &= E[v] + \Sigma_{vX}\Sigma_{XX}^{-1}(X - E[X]) = \frac{1}{1 + \delta^2}X \\ E[Z|X] &= E[Z] + \Sigma_{ZX}\Sigma_{XX}^{-1}(X - E[X]) = \frac{\delta}{1 + \delta^2}X \\ E[Y|Z] &= (\gamma + \beta\delta)Z \\ E[Y|X, Z] &= X\beta + Z\gamma \end{aligned} \tag{25}$$

In particular,

$$\begin{aligned}
E[Y|X] &= E[Y] + \Sigma_{YX}\Sigma_{XX}^{-1}(X - E[X]) = \Sigma_{YX}\Sigma_{XX}^{-1}X = \frac{\delta(\delta\beta + \gamma)\sigma_Z^2 + \beta\sigma_v^2}{\delta^2\sigma_Z^2 + \sigma_v^2}X \\
&= \left(\beta + \frac{\delta\gamma\sigma_Z^2}{\delta^2\sigma_Z^2 + \sigma_v^2}\right)X
\end{aligned} \tag{26}$$

$$E[\hat{\beta}|X] = \beta + \frac{\delta\gamma\sigma_Z^2}{\delta^2\sigma_Z^2 + \sigma_v^2} \tag{27}$$

This expression simplifies further when the variables are independent and identically distributed as a standard Normal, $(u, v, Z) \sim N[0, I]$, since

$$E[\hat{\beta}|X] = \beta + \frac{\gamma\delta}{1 + \delta^2} \tag{28}$$

When $\gamma = \delta = 0$, there is no confounder, thus $E[\hat{\beta}|X] = \beta$.

This completes the proof.

A.2. PROOF OF PROPOSITIONS IN SECTION 3.2

Consider a data-generating process with equations:

$$\begin{aligned}
Y &:= X\beta + u \\
Z &:= Y\gamma + X\delta + v
\end{aligned} \tag{29}$$

where $\gamma \neq 0$, $\delta \neq 0$, and variables (u, v, X) are independent and identically distributed as a Gaussian. The process is unknown to observers, who attempt to estimate the causal effect of X on Y by fitting the equation $Y = X\beta + Z\theta + \varepsilon$ on a sample produced by the process.

First, note that the equation for Z can be rewritten as

$$Z := (X\beta + u)\gamma + X\delta + v = X(\beta\gamma + \delta) + u\gamma + v \tag{30}$$

This can be written

$$\begin{bmatrix} X \\ u \\ v \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \beta & 1 & 0 \\ \delta + \beta\gamma & \gamma & 1 \end{bmatrix} \begin{bmatrix} X \\ u \\ v \end{bmatrix} = A \begin{bmatrix} X \\ u \\ v \end{bmatrix} \tag{31}$$

where $(X, u, v) \sim N(0, I)$. The variance is then

$$\text{Var} \begin{bmatrix} X \\ u \\ v \\ Y \\ Z \end{bmatrix} = AA' = \begin{bmatrix} 1 & 0 & 0 & \beta & \beta\gamma + \delta \\ 0 & 1 & 0 & 1 & \gamma \\ 0 & 0 & 1 & 0 & 1 \\ \beta & 1 & 0 & 1 + \beta^2 & \beta(\beta\gamma + \delta) + \gamma \\ \beta\gamma + \delta & \gamma & 1 & \beta(\beta\gamma + \delta) + \gamma & 1 + \gamma^2 + (\beta\gamma + \delta)^2 \end{bmatrix} \quad (32)$$

From this matrix, we can read the covariances between X, Y, Z, u, v . We can also use it to compute conditional expectations (Eaton [1983, pp. 116-117]), for instance,

$$\begin{aligned} E[Z|X] &= E[Z] + \Sigma_{ZX}\Sigma_{XX}^{-1}(X - E[X]) \\ &= (\beta\gamma + \delta)X \\ E[u|X, Z] &= E[u] + \Sigma_{u,XZ}\Sigma_{XZ,XZ}^{-2} \begin{pmatrix} X - E[X] \\ Z - E[Z] \end{pmatrix} \\ &= (0 \quad \gamma) \begin{pmatrix} 1 & \beta\gamma + \delta \\ \beta\gamma + \delta & 1 + \gamma^2 + (\beta\gamma + \delta)^2 \end{pmatrix}^{-1} \begin{pmatrix} X \\ Z \end{pmatrix} \\ &= -\frac{\gamma(\beta\gamma + \delta)}{1 + \gamma^2}X + \frac{\gamma}{1 + \gamma^2}Z \end{aligned} \quad (33)$$

In particular,

$$\begin{aligned} E[Y|X, Z] &= E[Y] + \Sigma_{Y,XZ}\Sigma_{XZ,XZ}^{-1} \begin{pmatrix} X \\ Z \end{pmatrix} - E \begin{pmatrix} X \\ Z \end{pmatrix} \\ &= [\Sigma_{YX} \quad \Sigma_{YZ}] \begin{bmatrix} \Sigma_{XX} & \Sigma_{XZ} \\ \Sigma_{ZX} & \Sigma_{ZZ} \end{bmatrix}^{-1} \begin{pmatrix} X \\ Z \end{pmatrix} \\ &= \frac{1}{\Sigma_{XX}\Sigma_{ZZ} - \Sigma_{XZ}^2} [\Sigma_{YX} \quad \Sigma_{YZ}] \begin{bmatrix} \Sigma_{ZZ} & -\Sigma_{XZ} \\ -\Sigma_{ZX} & \Sigma_{XX} \end{bmatrix} \begin{pmatrix} X \\ Z \end{pmatrix} \\ &= \frac{1}{\Sigma_{XX}\Sigma_{ZZ} - \Sigma_{XZ}^2} [\Sigma_{YX}\Sigma_{ZZ} - \Sigma_{YZ}\Sigma_{ZX} \quad \Sigma_{YZ}\Sigma_{XX} - \Sigma_{YX}\Sigma_{XZ}] \begin{pmatrix} X \\ Z \end{pmatrix} \end{aligned} \quad (34)$$

From the above, we obtain that

$$E[\hat{\beta}|X, Z] = \frac{\Sigma_{YX}\Sigma_{ZZ} - \Sigma_{YZ}\Sigma_{ZX}}{\Sigma_{XX}\Sigma_{ZZ} - \Sigma_{XZ}^2} = \frac{\beta\sigma_v^2 - \delta\gamma\sigma_u^2}{\sigma_v^2 + \gamma^2\sigma_u^2} \quad (35)$$

$$E[\hat{\theta}|X, Z] = \frac{\Sigma_{YZ}\Sigma_{XX} - \Sigma_{YX}\Sigma_{XZ}}{\Sigma_{XX}\Sigma_{ZZ} - \Sigma_{XZ}^2} = \frac{\gamma\sigma_u^2}{\sigma_v^2 + \gamma^2\sigma_u^2} \quad (36)$$

This expression simplifies further when the variables are independent and identically distributed as a standard Normal, $(u, v, X) \sim N[0, I]$, since

$$E[\hat{\beta}|X, Z] = \frac{\beta - \delta\gamma}{1 + \gamma^2} \quad (37)$$

$$E[\hat{\theta}|X, Z] = \frac{\gamma}{1 + \gamma^2} \quad (38)$$

When $\gamma = \delta = 0$, there is no collider, thus $E[\hat{\beta}|X, Z] = \beta$ and $E[\hat{\theta}|X, Z] = 0$.

This completes the proof.

A.3. PROOFS OF PROPOSITIONS IN SECTION 4

Consider the following collider model, where $\gamma \neq 0$, $\delta \neq 0$, and $(X, u, v) \sim N[0, I]$:

$$\begin{aligned} Y &:= X\beta + u \\ Z &:= Y\gamma + X\delta + v \end{aligned} \quad (39)$$

We can rewrite Z as

$$Z := (X\beta + u)\gamma + X\delta + v = X(\beta\gamma + \delta) + u\gamma + v \quad (40)$$

The model that over-controls for a collider is

$$\begin{aligned} Y &= XE[\hat{\beta}|X, Z] + ZE[\hat{\theta}|X, Z] + \varepsilon \\ &= \frac{\beta - \delta\gamma}{1 + \gamma^2}X + \frac{\gamma}{1 + \gamma^2}Z + \varepsilon \end{aligned} \quad (41)$$

The residuals of the over-controlled model are

$$\begin{aligned} \varepsilon &= Y - XE[\hat{\beta}|X, Z] - ZE[\hat{\theta}|X, Z] \\ &= (X\beta + u) - XE[\hat{\beta}|X, Z] - [X(\beta\gamma + \delta) + u\gamma + v]E[\hat{\theta}|X, Z] \\ &= X[\beta - E[\hat{\beta}|X, Z] - E[\hat{\theta}|X, Z](\beta\gamma + \delta)] + u(1 - E[\hat{\theta}|X, Z]\gamma) - E[\hat{\theta}|X, Z]v \\ &= X\left[\beta - \frac{\beta - \delta\gamma}{1 + \gamma^2} - \frac{\gamma}{1 + \gamma^2}(\beta\gamma + \delta)\right] + u\left[1 - \frac{\gamma}{1 + \gamma^2}\gamma\right] - \frac{\gamma}{1 + \gamma^2}v \\ &= X\frac{\beta(1 + \gamma^2) - (\beta - \delta\gamma) - \gamma(\beta\gamma + \delta)}{1 + \gamma^2} + u\frac{1}{1 + \gamma^2} - \frac{\gamma}{1 + \gamma^2}v \\ &= \frac{1}{1 + \gamma^2}u - \frac{\gamma}{1 + \gamma^2}v \end{aligned} \quad (42)$$

The variances of ε and Y in the over-controlled model are

$$Var[\varepsilon] = \left(\frac{1}{1 + \gamma^2}\right)^2 + \left(\frac{\gamma}{1 + \gamma^2}\right)^2 = \frac{1 + \gamma^2}{(1 + \gamma^2)^2} = \frac{1}{1 + \gamma^2} \quad (43)$$

$$Var[Y] = \beta^2 + 1 \quad (44)$$

A.3.1. R-SQUARED

The R-squared (R^2) of the correctly specified model (if the estimated coefficients have their expected values) is

$$\begin{aligned} R^2 &= 1 - \frac{\text{Var}[u]}{\text{Var}[Y]} \\ &= 1 - \frac{1}{1 + \beta^2} \end{aligned} \tag{45}$$

For the over-controlled model, the R^2 is

$$\begin{aligned} R^2 &= 1 - \frac{\text{Var}[\varepsilon]}{\text{Var}[Y]} \\ &= 1 - \frac{1}{(1 + \gamma^2)(1 + \beta^2)} \end{aligned} \tag{46}$$

Therefore, for R^2 we always have:

$$R^2_{\text{over-controlled}} > R^2_{\text{correct}} \tag{47}$$

This completes the proof.

A.3.2. ADJUSTED R-SQUARED

Since the number of estimated parameters in the correct model is $k = 1$, the adjusted R^2 for T observations is

$$\begin{aligned} R^2_{\text{adj}} &= 1 - \frac{\text{Var}[u]}{\text{Var}[Y]} \frac{T - 1}{T - k - 1} \\ &= 1 - \frac{1}{1 + \beta^2} \frac{T - 1}{T - 2} \end{aligned} \tag{48}$$

Since we now have $k = 2$ estimated parameters, the adjusted R^2 of the over-controlled model is

$$\begin{aligned} R^2_{\text{adj}} &= 1 - \frac{\text{Var}[\varepsilon]}{\text{Var}[Y]} \frac{T - 1}{T - k - 1} \\ &= 1 - \frac{1}{(1 + \gamma^2)(1 + \beta^2)} \frac{T - 1}{T - 3} \end{aligned} \tag{49}$$

Therefore, if $|\gamma|$ is sufficiently large relative to the number of observations, for the adjusted R^2 we have:

$$\begin{aligned}
R_{\text{adj, over-controlled}}^2 > R_{\text{adj, correct}}^2 &\Leftrightarrow \frac{1}{(1 + \gamma^2)(1 + \beta^2)} \frac{T - 1}{T - 3} < \frac{1}{1 + \beta^2} \frac{T - 1}{T - 2} \\
&\Leftrightarrow \frac{1}{1 + \gamma^2} < \frac{T - 3}{T - 2} \\
&\Leftrightarrow 1 + \gamma^2 > \frac{T - 2}{T - 3} \\
&\Leftrightarrow |\gamma| > \frac{1}{\sqrt{T - 3}}
\end{aligned} \tag{50}$$

This completes the proof.

A.3.3. P-VALUES

For a linear regression model $Y = X\beta + Z\theta + \varepsilon$, fitted on T observations, the covariance matrix of the coefficients is

$$\begin{aligned}
\text{Var} \begin{bmatrix} E[\hat{\beta}|X, Z] \\ E[\hat{\theta}|X, Z] \end{bmatrix} &= \text{Var}[\varepsilon] \begin{bmatrix} T \text{Var}[X] & T \text{Cov}[X, Z] \\ T \text{Cov}[X, Z] & T \text{Var}[Z] \end{bmatrix}^{-1} \\
&= \frac{1}{T} \frac{\text{Var}[\varepsilon]}{\text{Var}[X] \text{Var}[Z] - \text{Cov}[X, Z]^2} \begin{bmatrix} \text{Var}[Z] & -\text{Cov}[X, Z] \\ -\text{Cov}[X, Z] & \text{Var}[X] \end{bmatrix}
\end{aligned} \tag{51}$$

For individual tests on the regression coefficients, we only need the diagonal elements. In the case where the model over-controls for a collider, the variances of the estimated coefficients are:

$$\begin{aligned}
\text{Var} [E[\hat{\beta}|X, Z]] &= \frac{\text{Var}[\varepsilon]}{T} \frac{\text{Var}[Z]}{\text{Var}[X] \text{Var}[Z] - \text{Cov}[X, Z]^2} \\
&= \frac{1}{T} \frac{1}{1 + \gamma^2} \frac{(\beta\gamma + \delta)^2 + \gamma^2 + 1}{(\beta\gamma + \delta)^2 + \gamma^2 + 1 - (\beta\gamma + \delta)^2} \\
&= \frac{(\beta\gamma + \delta)^2 + \gamma^2 + 1}{T(1 + \gamma^2)^2}
\end{aligned} \tag{52}$$

$$\begin{aligned}
\text{Var} [E[\hat{\theta}|X, Z]] &= \frac{\text{Var}[\varepsilon]}{T} \frac{\text{Var}[X]}{\text{Var}[X] \text{Var}[Z] - \text{Cov}[X, Z]^2} = \frac{1}{T} \frac{1}{1 + \gamma^2} \frac{1}{1 + \gamma^2} \\
&= \frac{1}{T(1 + \gamma^2)^2}
\end{aligned} \tag{53}$$

Under the null hypothesis, the t-values of the estimated coefficients in the over-controlled model follow a t-distribution with $T - 2$ degrees of freedom,

$$\begin{aligned}
t_{E[\hat{\beta}|X,Z]} &= \frac{E[\hat{\beta}|X,Z]}{\sqrt{\text{Var}[E[\hat{\beta}|X,Z]]}} = \frac{\frac{\beta - \delta\gamma}{1 + \gamma^2}}{\sqrt{\frac{(\beta\gamma + \delta)^2 + \gamma^2 + 1}{T(1 + \gamma^2)^2}}} \\
&= \sqrt{T} \frac{\beta - \delta\gamma}{\sqrt{(\beta\gamma + \delta)^2 + \gamma^2 + 1}}
\end{aligned} \tag{54}$$

$$t_{E[\hat{\theta}|X,Z]} = \frac{E[\hat{\theta}|X,Z]}{\sqrt{\text{Var}[E[\hat{\theta}|X,Z]]}} = \frac{\frac{\gamma}{1 + \gamma^2}}{\frac{1}{\sqrt{T(1 + \gamma^2)^2}}} = \sqrt{T}\gamma \tag{55}$$

For the correctly specified model, the computations are simpler. The variance of the coefficient is

$$\text{Var}[E[\hat{\beta}|X,Z]] = \text{Var}[u](T \text{Var}[X])^{-1} = \frac{1}{T} \tag{56}$$

Under the null hypothesis, the t-value of the correctly specified model follows a t-distribution with $T - 1$ degrees of freedom,

$$t_{E[\hat{\beta}|X,Z]} = \frac{E[\hat{\beta}|X,Z]}{\sqrt{\text{Var}[E[\hat{\beta}|X,Z]]}} = \frac{E[\hat{\beta}|X,Z]}{\sqrt{1/T}} = \sqrt{T}\beta \tag{57}$$

In conclusion, false positives become more likely in models that over-control for a collider, because: (a) $E[\hat{\theta}|X,Z]$ is statistically significant for $T \gg 0$; and (b) for some values of δ, γ (in particular, $|\delta| \gg 0, |\gamma| \gg 0, |\delta| \approx |\gamma|$), or when the causal effect is weak ($|\beta| \approx 0$), then

$$|t_{E[\hat{\beta}|X,Z], \text{over-controlled}}| > |t_{E[\hat{\beta}|X,Z], \text{correct}}| \tag{58}$$

This completes the proof.

A.4. CONSTANT RISK PREMIA WITH CONFOUNDING EFFECT SHIFT

Consider the data-generating process described in Section 3.1. After a sufficient sample size has been generated, an investor can estimate parameters with sufficient confidence, and design an investment strategy. Suppose that, sometime afterwards, the confounding effect shifts from δ_0 to δ_1 , while keeping constant the risk premia (β, γ) . For a long period of time, there are not enough observations to detect that shift, thus the investor continues to use the original parameter estimates. Next, we study whether that shift in $\delta_0 \neq \delta_1$, while keeping constant the risk premia (β, γ) , can be enough to cause systematic losses.

A.4.1. FACTOR STRATEGIES

After the shift in parameters, a factor strategy based on the correctly specified model that sizes positions by $E_0[\hat{Y}|X, Z]$ delivers expected returns

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X, Z]|X, Z] &= E_1[Y(X\beta + Z\gamma)|X, Z] \\
&= E_1[Y|X, Z](X\beta + Z\gamma) \\
&= E_1[X\beta + Z\gamma + u|X, Z](X\beta + Z\gamma) \\
&= (X\beta + Z\gamma)^2 \geq 0
\end{aligned} \tag{59}$$

This equation does not include δ_1 , therefore the performance of the factor strategy based on the correctly specified model is not affected by the change in $\delta_0 \neq \delta_1$.

In contrast, for a factor strategy based on a model that under-controls for a confounder and sizes positions by $E_0[\hat{Y}|X]$, the expected returns after the parameter shift are

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X]|X] &= E_1\left[Y\left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2}\right)X|X\right] \\
&= E_1[Y|X]\left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2}\right)X \\
&= X^2\left(\beta + \frac{\gamma\delta_1}{1 + \delta_1^2}\right)\left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2}\right)
\end{aligned} \tag{60}$$

For appropriate combinations of real values of the above variables, the strategy yields systematic losses, i.e. $E_1[YE_0[\hat{Y}|X]|X] < 0$. For example, the strategy yields systematic losses under $(\beta, \gamma, \delta_0, \delta_1) = (1, 3, 1, -1)$, even though the risk premia remain constant and $E_0[\hat{\beta}|X] = 2.5$ is estimated with the correct sign ($\beta = 1$).

A.4.2. FORECASTING STRATEGIES

After the shift in parameters, a forecasting strategy based on the correctly specified model that sizes positions by $E_0[\hat{Y}|X, Z]$ delivers expected returns

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X, Z]] &= E_1[Y(X\beta + Z\gamma)] \\
&= \beta E_1[XY] + \gamma E_1[YZ] \\
&= \beta(\beta(1 + \delta_1^2) + \delta_1\gamma) + \gamma(\gamma + \beta\delta_1) \\
&= \beta^2 + (\beta\delta_1 + \gamma)^2 \geq 0
\end{aligned} \tag{61}$$

This equation does include δ_1 , therefore the performance of the forecasting strategy based on the correctly specified model is affected by the change in $\delta_0 \neq \delta_1$, however not to the point of experiencing systematic losses.

In contrast, for a forecasting strategy based on a model that under-controls for a confounder and sizes positions by $E_0[\hat{Y}|X]$, the expected returns after the parameter shift are

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X]] &= E_1[YX] \left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2} \right) \\
&= (\beta(1 + \delta_1^2) + \gamma\delta_1) \left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2} \right) \\
&= (1 + \delta_1^2) \left(\beta + \frac{\gamma\delta_1}{1 + \delta_1^2} \right) \left(\beta + \frac{\gamma\delta_0}{1 + \delta_0^2} \right)
\end{aligned} \tag{62}$$

For appropriate combinations of real values of the above variables, the strategy yields systematic losses, i.e. $E_1[YE_0[\hat{Y}|X]] < 0$. For example, the strategy yields systematic losses under $(\beta, \gamma, \delta_0, \delta_1) = (1, 3, 1, -1)$, even though the risk premia remain constant and $E_0[\hat{\beta}|X] = 2.5$ is estimated with the correct sign ($\beta = 1$).

From the above results, we conclude that a shift in the confounding effect δ : (a) can cause systematic losses in factor and forecasting strategies that fail to control for a collider; (b) does not affect the performance of correctly specified factor strategies, which remains non-negative; and (c) affects the performance of correctly specified factor strategies, however not to the point of yielding systematic losses.

A.5. CONSTANT RISK PREMIUM WITH COLLIDER PARAMETERS SHIFT

Consider the data-generating process described in Section 3.2. After a sufficient sample size has been generated, an investor can estimate parameters with sufficient confidence, and design an investment strategy. Suppose that, some time afterwards, the collider parameters shift from (γ_0, δ_0) to (γ_1, δ_1) , while keeping constant the risk premium β . For a long period of time, there are not enough observations to detect that shift, thus the investor continues to use the original parameter estimates. Next, we study whether that shift in $\gamma_0 \neq \gamma_1$ or $\delta_0 \neq \delta_1$, while keeping constant the only risk premium β , can be enough to cause systematic losses.¹²

A.5.1. FACTOR STRATEGIES

After the shift in parameters, a factor strategy based on the correctly specified model that sizes positions by $E_0[\hat{Y}|X]$ delivers expected returns

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X]|X] &= E_1[Y|X]X\beta \\
&= E_1[X\beta + u|X]X\beta \\
&= (X\beta)^2 \geq 0
\end{aligned} \tag{63}$$

This equation does not include γ_1 or δ_1 , therefore the shift in collider parameters has no effect on the performance of a factor strategy based on the correctly specified model. Intuitively, the performance of the correctly specified strategy is unaffected by a shift in collider parameters because: (a) the collider parameters have no effect on neither X or Y ; and (b) the estimates of the correctly specified model are not influenced by the collider parameters.

¹² Strictly speaking, a shift in the confounding effect or collider parameters could increase profitability, however investors are generally more concerned about unforeseen losses than they are about unforeseen profits.

In contrast, for a factor strategy based on a model that over-controls for a collider and sizes positions by $E_0[\hat{Y}|X, \tilde{Z}]$, the expected returns after the parameter shift are

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X, \tilde{Z}]|X, \tilde{Z}] &= E_1[(X\beta + u)(XE_0[\hat{\beta}|X, \tilde{Z}] + \tilde{Z}E_0[\hat{\theta}|X, \tilde{Z}])|X, \tilde{Z}] \\
&= (X\beta + E_1[u|X, \tilde{Z}]) (XE_0[\hat{\beta}|X, \tilde{Z}] + \tilde{Z}E_0[\hat{\theta}|X, \tilde{Z}]) \quad (64) \\
&= X\beta \frac{X\beta + \gamma_0(\tilde{Z} - \delta_0 X)}{1 + \gamma_0^2}
\end{aligned}$$

This is the same result that was derived under constant correlation (Section 3.2.1). The reason is, a change in the collider parameters affects Z , but it has no effect on $\tilde{Z}$. Like before, for appropriate combinations of real values of the above variables, the strategy yields systematic losses, i.e. $E_1[YE_0[\hat{Y}|X, \tilde{Z}]|X, \tilde{Z}] < 0$.

A.5.2. FORECASTING STRATEGIES

After the shift in parameters, a forecasting strategy based on the correctly specified model that sizes positions by $E_0[\hat{Y}|X]$ delivers expected returns

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X]] &= E_1[(\beta X + u)\beta X] \\
&= \beta^2 E_1[X^2] + \beta E_1[uX] \\
&= \beta^2 \geq 0 \quad (65)
\end{aligned}$$

This equation does not include γ_1 or δ_1 , therefore the shift in collider parameters has no effect on the performance of a forecasting strategy based on the correctly specified model.

However, for a forecasting strategy based on a model that over-controls for a collider and sizes positions by $E_0[\hat{Y}|X, \tilde{Z}]$, the expected returns after the parameter shift are

$$\begin{aligned}
E_1[YE_0[\hat{Y}|X, \tilde{Z}]] &= E_1[Y(XE_0[\hat{\beta}|X, \tilde{Z}] + \tilde{Z}E_0[\hat{\theta}|X, \tilde{Z}])] \\
&= E_0[\hat{\beta}|X, \tilde{Z}]E_1[XY] + E_0[\hat{\theta}|X, \tilde{Z}]E_1[Y\tilde{Z}] \\
&= \beta \frac{\beta - \gamma_0\delta_0}{1 + \gamma_0^2} \quad (66)
\end{aligned}$$

This is the same result that was derived under constant correlation (Section 3.2.2). The reason is, a change in the collider parameters affects Z , but it has no effect on $\tilde{Z}$. Like before, for appropriate combinations of real values of the above variables, the strategy yields systematic losses, i.e. $E_1[YE_0[\hat{Y}|X, \tilde{Z}]] < 0$.

From the above results, we conclude that a shift in the collider parameters (γ, δ) : (a) does not affect the performance of strategies based on the correctly specified model, whether factor strategies or forecasting strategies, which remains non-negative; and (b) does not affect the performance of strategies based on a model that over-controls for a collider, whether factor strategies or forecasting strategies, which can be negative. The intuition behind this conclusion is

that a change in collider parameters has no effect on Y , hence it does not impact the strategy's performance, unlike in the confounder case.

A.6. GLOSSARY OF TECHNICAL TERMS

This glossary defines technical terms used throughout the article that may not be familiar to all readers, particularly those related to causal inference, machine learning, and model evaluation.

Backdoor Adjustment — A causal inference technique used to block all non-causal paths (backdoor paths) between treatment and outcome by conditioning on confounders, thus identifying causal effects.

Berkson's Fallacy — A statistical bias that arises when conditioning on a collider, which induces a spurious association between otherwise independent variables.

Binary Classification Scores — Metrics used to evaluate the quality of a model's probabilistic or categorical predictions for binary outcomes. Common scores include:

- **Brier Score**: Measures the mean squared difference between predicted probabilities and actual binary outcomes. Lower values indicate better calibration and accuracy.
- **Log-Loss (Logarithmic Loss)**: Also known as cross-entropy loss, it penalizes incorrect predictions more heavily when made with high confidence. Lower values indicate better calibration and accuracy.
- **Receiver Operating Characteristic (ROC) Curve**: Plots the trade-off between true positive rate and false positive rate. The Area Under the Curve (AUC) summarizes performance. It can overstate the goodness of the classifier when the classes are highly imbalanced.
- **Precision-Recall (PR) Curve**: Plots precision versus recall across thresholds. The Area Under the Curve (AUC) summarizes performance. It is especially useful for imbalanced classes.
- **Mean Reciprocal Rank (MRR)**: Evaluates how high the first relevant result appears in a probability-ranked list of predictions.
- **Classification Accuracy**: The ratio of correct predictions to total predictions. While intuitive, it can be misleading for imbalanced datasets.

Causal Adjustment Set — A set of control variables that block all spurious (non-causal) paths without introducing collider or mediator bias. An adjustment set is not necessarily unique; multiple valid sets may achieve the same goal.

Causal Discovery — A family of algorithms that infer the structure of causal relationships from observational data, under assumptions such as acyclicity (e.g., PC algorithm, LiNGAM).

Causal Graph — A directed acyclic graph (DAG) representing hypothesized causal relationships among variables. Used to determine valid control strategies and assess the correct specification of the regression model.

Causal Inference — Algorithms or models using a causal graph to assess the causal effect of one variable on another. Examples include backdoor adjustment, front-door adjustment, instrumental variables, inverse propensity weighting, doubly robust models, double machine learning, etc.

Classification Accuracy — See Binary Classification Scores.

Collider — A variable that is causally influenced by variables on the left- and right-hand side of the regression equation. Conditioning on a collider opens a non-causal path, introducing spurious associations and estimation bias.

Confounder — A variable that causally influences variables on the left- and right-hand side of the regression equation. Failure to control for a confounder leaves open a non-causal path, introducing spurious associations and estimation bias.

Deflated Sharpe Ratio (DSR) — A probabilistic Sharpe ratio adjusted for selection bias under multiple testing, sample length, skewness and kurtosis, reflecting the true likelihood that a strategy's performance is not due to chance.

Do-Calculus — A set of three algebraic rules for estimating the causal effect of one variable on another, from observational data alone, assuming that the causal graph is known, by simulating a controlled experiment.

Embargo (in cross-validation) — A time buffer applied after a test window, within which training data is dropped to prevent time-series leakage and autocorrelation bias.

False Strategy Theorem — A mathematical result proving that repeated backtesting increases the likelihood of overfitting, such that with enough trials any Sharpe ratio is achievable in-sample, even if the true Sharpe ratio is zero.

Front-door Adjustment — A causal identification strategy used when the treatment affects an intermediate cause (mediator), which in turn affects the outcome, under specific conditions. Useful when there are known but unobserved confounders.

GES algorithm — A causal Discovery algorithm based on goodness-of-fit tests.

Instrumental Variable — A variable that influences the treatment but affects the outcome only through its effect on the treatment. It must not be correlated with unobserved confounders of the treatment–outcome relationship. Useful when there are known but unobserved confounders.

Kitchen Sink Regression — A modeling approach where all available variables are included without theoretical justification. While it may improve statistical fit, it increases the risk of overfitting and causal misspecification, particularly through the inclusion of colliders.

LiNGAM (Linear Non-Gaussian Acyclic Model) — A causal discovery algorithm that identifies causal directions under the assumptions of linearity and non-Gaussian error terms.

Log-Loss (Logarithmic Loss) — See Binary Classification Scores.

Mean Reciprocal Rank (MRR) — See Binary Classification Scores.

Mean Squared Error (MSE) — The average of squared differences between predicted and actual values in regression problems. Lower values indicate better predictive accuracy.

Micro, Macro, and Weighted Averaging (of classification scores) — Methods for aggregating binary performance metrics across multiple classes:

- **Micro averaging:** Aggregates globally, favoring frequent classes. It answers the question: “What is the performance of the classifier on a randomly chosen observation?” It is preferable when all predictions are equally important.
- **Macro averaging:** Computes metrics per class and averages equally. It answers the question: “What is the performance of the classifier on a randomly chosen class?” It is preferable when all classes are equally important.
- **Weighted averaging:** Averages metrics per class weighted by class frequency. It answers the question: “What is the performance of the classifier on the class of a randomly chosen observation?” It strikes a balance between performance across all observations and performance across all classes.

One-vs-Rest (OvR) Encoding — A strategy to generalize binary classification methods to multi-class problems by training one classifier per class, versus all others (also known as one-hot encoding).

Over-Control — The inclusion of unnecessary control variables in a regression model. When the over-controlled variable is a collider or a mediator, it introduces estimation bias and undermines causal interpretability.

Overlapping Returns — Return series where the same price observations are reused across multiple periods, leading to biased estimates if not corrected with techniques like purged cross-validation.

PC Algorithm — A widely used causal discovery method that relies on conditional independence testing to estimate a partially directed acyclic graph from data.

Permutation Feature Importance (PFI) — A model-agnostic method for quantifying feature importance by measuring the change in a chosen performance metric (e.g., accuracy, AUC, MSE) after permuting each feature. A larger performance drop indicates higher importance. PFI generalizes Mean Decrease Accuracy (MDA) to arbitrary metrics, and is sensitive to feature correlation and model variance.

P-Hacking — The malpractice of manipulating model specifications or repeated testing to produce statistically significant results, often resulting in false discoveries.

Precision-Recall (PR) Curve — See Binary Classification Scores.

Purged Cross-Validation — A cross-validation procedure that removes overlapping observations between training and test sets to eliminate lookahead bias in temporally dependent data.

Receiver Operating Characteristic (ROC) Curve — See Binary Classification Scores.

Reverse Causation — A modeling error where the direction of causality is incorrectly assumed, leading to biased inference. Occurs when the outcome influences the predictor rather than vice versa.

R-Squared (R^2) — The proportion of variance in the dependent variable explained by the model. Commonly used for explanatory power in regression. The adjusted R^2 regularizes the R^2 by taking into account the number of explanatory variables.

Shapley Values — A method from cooperative game theory used to quantify the marginal contribution of each input variable to a model's output, ensuring fairness in feature attribution.

Sharpe Ratio — A common risk-adjusted return metric, calculated as the average excess return divided by the standard deviation of excess return.

Simpson's Paradox — A statistical phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. This occurs due to the presence of confounding variables that are not properly accounted for, potentially leading to misleading conclusions about causal relationships.

Specification Error — A form of model misspecification caused by omitting relevant variables (e.g., confounders) or including inappropriate ones (e.g., colliders), resulting in biased estimates.

Spearman Correlation — A non-parametric measure of rank correlation that assesses how well the relationship between two variables can be described using a monotonic function.

Statistical Arbitrage — A class of trading strategies that aim to profit from statistical mispricings in asset prices, often using mean-reverting techniques like pairs trading.

Systematic Losses — Persistent, non-random investment losses caused by structural flaws in a strategy, such as misspecified factor models.

Systematic Risk Premium — The expected return earned by bearing a non-diversifiable risk factor, assumed to persist across time and securities.

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